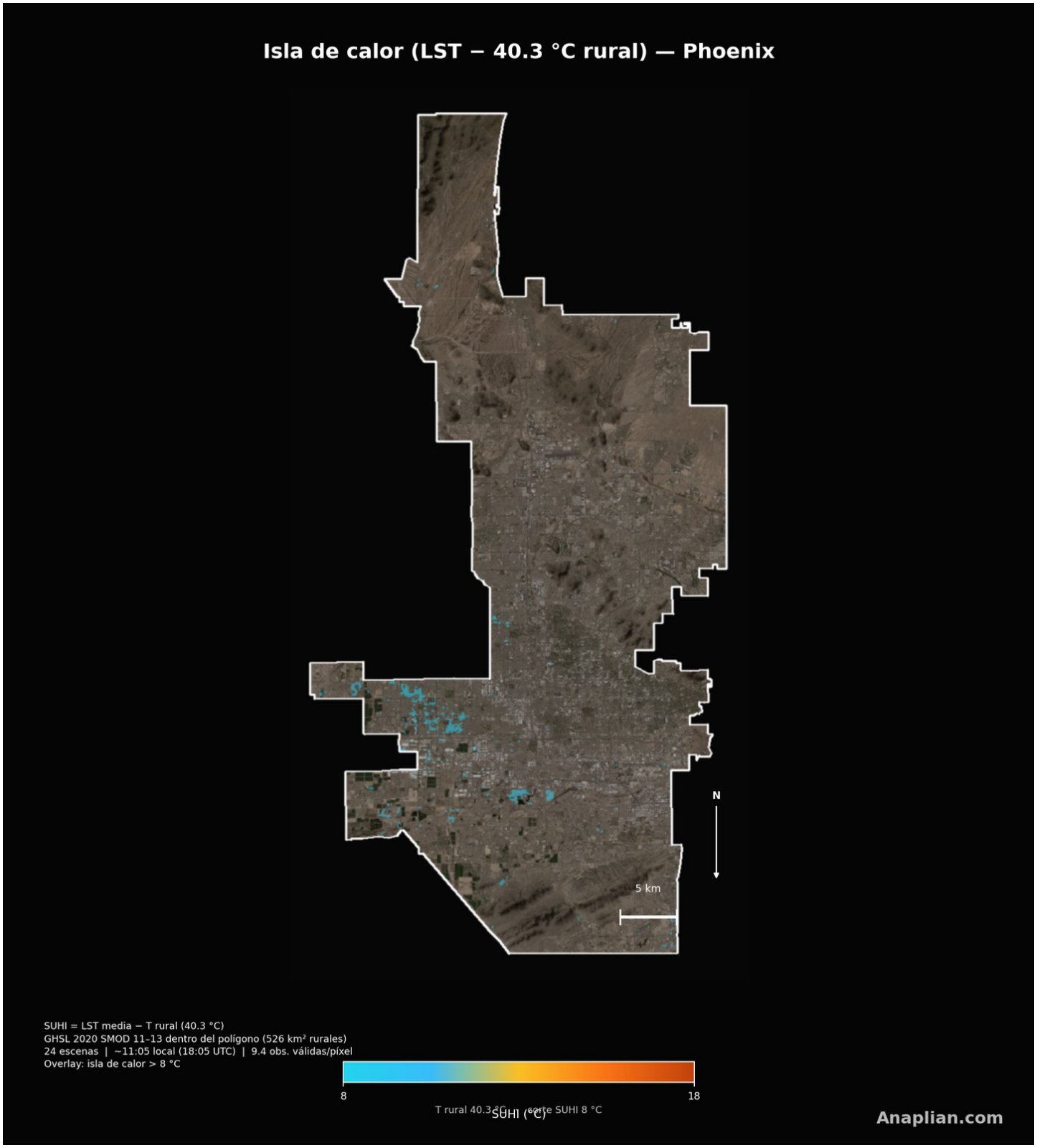
Climate Vulnerability Index for Phoenix



Surface heat island and urban-village boundaries.

Contents

1. Introduction
2. Position in the index
3. Highest-IVC subdivisions
4. Per-pixel maximum LST > 43 °C
5. Mean LST of observations > 35 °C
6. Vegetation (NDVI)
7. Heat island (pixel − rural)
8. Four sub-index elements
9. Table of urban villages
10. Urban villages
11. Why Phoenix is in Priority 2

1. Introduction

The sustained rise in mean temperature and in its variability — more extreme days, longer heatwaves — falls on human settlements. The body sheds heat by radiation, convection and, above all, evaporation of sweat. When air and radiating surfaces block that loss, core temperature rises, the heart works harder to send blood to the skin, and water and electrolytes are lost. The outcome can be heat exhaustion, heat stroke or cardiovascular and renal strain. Night-time heat, typical of urban islands, cuts recovery during sleep. The risk is higher for older people, those with chronic illness, outdoor workers and residents of poorly ventilated or unshaded housing.

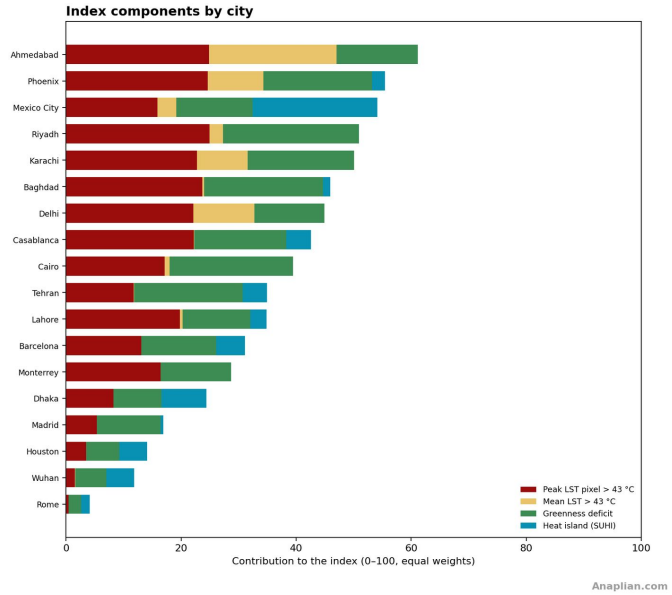
The purpose of this index is to quantify that surface heat load and make it comparable, so as to see where vulnerability concentrates and which urban traits foster it: pavement, sheds, vegetation deficit and heat islands relative to the surroundings. It does not replace demography or epidemiological surveillance; it ranks territory by surface exposure, with a fixed metric across cities.

The comparison works at two scales. Across cities, the same number sizes the problem against places that already apply good practice or illustrate solutions to heat concentration. Inside each city, the political units — in Phoenix, the urban villages — locate the most problematic zones in more detail and point to authorities and residents with a direct stake in addressing them.

For data availability and comparability we use four Landsat layers. Three are surface temperature: the period maximum, the mean of the observations and the heat island, the differential between a nearby rural or less-urbanised reference zone and the temperature in the area. The fourth is a vegetation index (NDVI) that enters inverted: lower greenness is penalised, because vegetation has a clear cooling effect.

2. Position in the index

Phoenix ranks 2 of 18 cities measured in April and May 2026, with a Climate Vulnerability Index (IVC) of 55.4 out of 100. Ahmedabad leads. Mexico City (54.1) comes next. Almost all of the urban pixel-max exceeds 43 °C (99 %); the composite mean does so on 39 % of urban land. Rural desert is already about 40 °C, so urban SUHI is only +1.3 °C. What sets Phoenix apart is pavement and a greenness deficit on a hot desert, not a CDMX-style island over a cool forest.



Index components by city.

Inside the city the index concentrates in the west and the core. Maryvale leads (IVC 71): arterials, parking and low roofs. Central City (71) adds downtown, warehouses and the Sky Harbor edge (NDVI 0.11). Estrella (65) is new sprawl on scraped desert. South Mountain (62) mixes the preserve with paved foothills. Ahwatukee Foothills (62) is master-planned tile roofs. Desert View ranks last (44), on the northern conservation fringe.

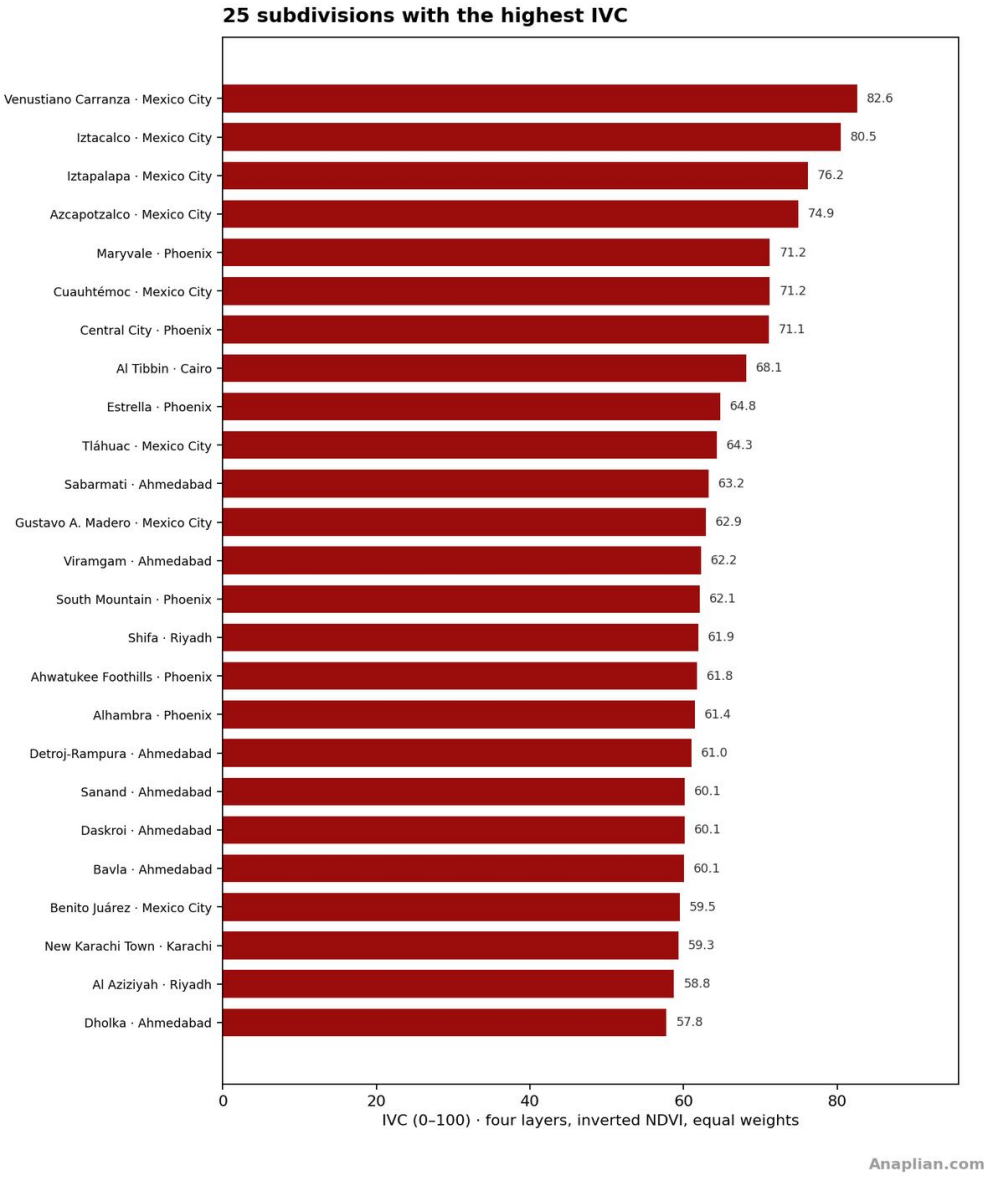
West-side villages (Maryvale, Central City) combine slab, parking and almost no canopy; mean LST still exceeds 43 °C on most of the urban mask. Northern villages (Desert View, North Gateway, Paradise Valley) keep a high pixel-max — desert roofs in April — but the mean falls because density and asphalt drop.

Unlike Mexico City, Phoenix cannot lean on a cool rural reference (~40 °C). Unlike Ahmedabad, the mean is not already saturated at 89 %. Design still moves the mean: shade, albedo and less parking separate Maryvale from Desert View. Desert View shows the floor of the city ranking.

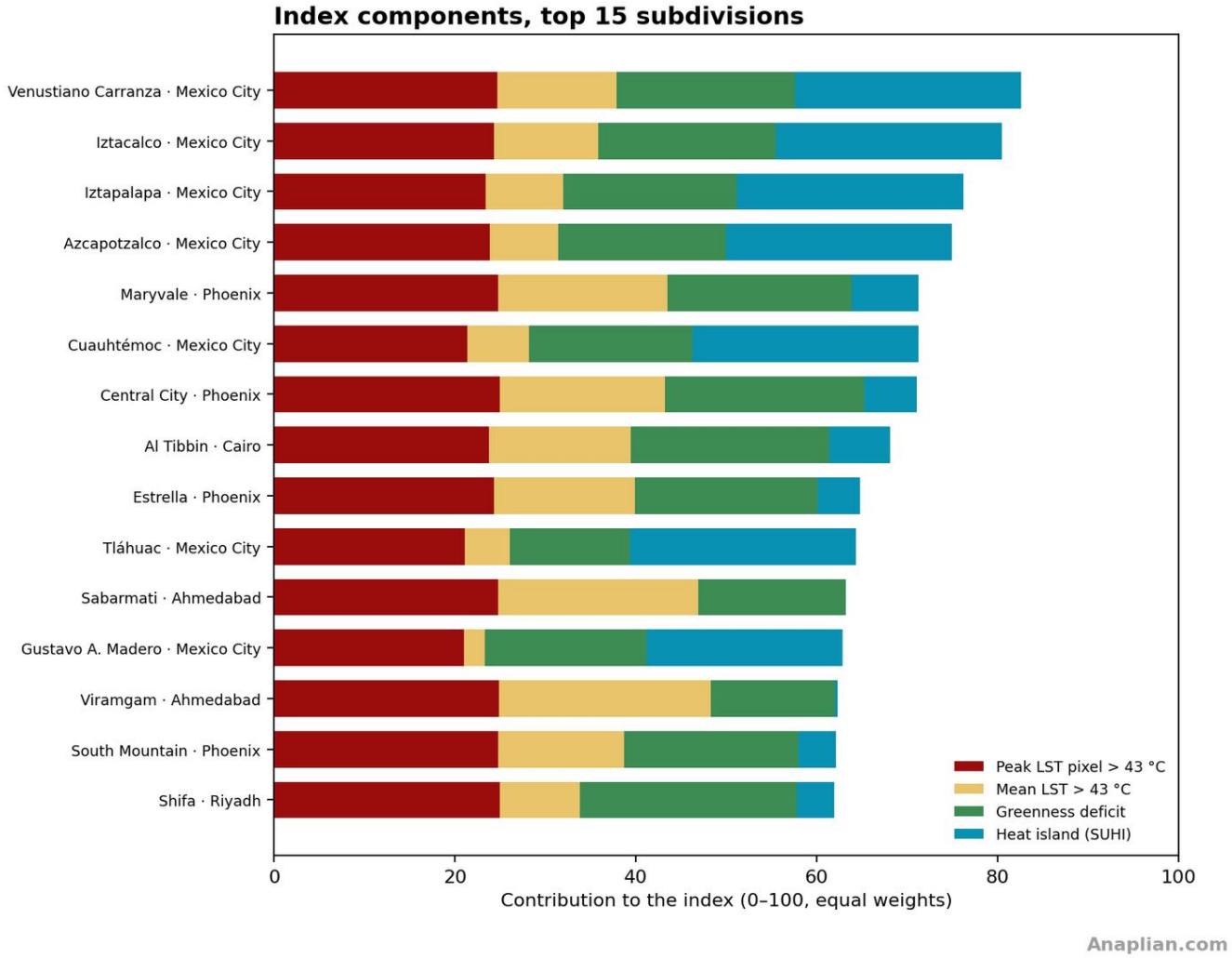
Land cover in the five highest-IVC urban villages is asphalt, parking, low roofs and airport or arterial slab, with little green. Maryvale is strip-mall parking and six-lane roads. Central City adds downtown decks and the Sky Harbor runways and aprons. Estrella is fresh pavement on desert. South Mountain’s urban score sits on the asphalted foothills, not on the preserve. Ahwatukee Foothills is tile roofs and arterials on desert lots. Canopy is fragmented; it does not offset the slab.

3. Highest-IVC subdivisions

Among 296 political subdivisions of the 18 cities in the comparable index, 6 Phoenix urban villages enter the top 25: Maryvale (5th, IVC 71), Central City (7th, IVC 71), Estrella (9th, IVC 65), South Mountain (14th, IVC 62), Ahwatukee Foothills (16th, IVC 62), Alhambra (17th, IVC 61). Maryvale sits 5th overall, between Azcapotzalco and Cuauhtémoc. The components chart shows high z_LST max and greenness deficit, with a modest SUHI: desert pavement, not a 13 °C island. Desert View (44) stays well outside the top 25.



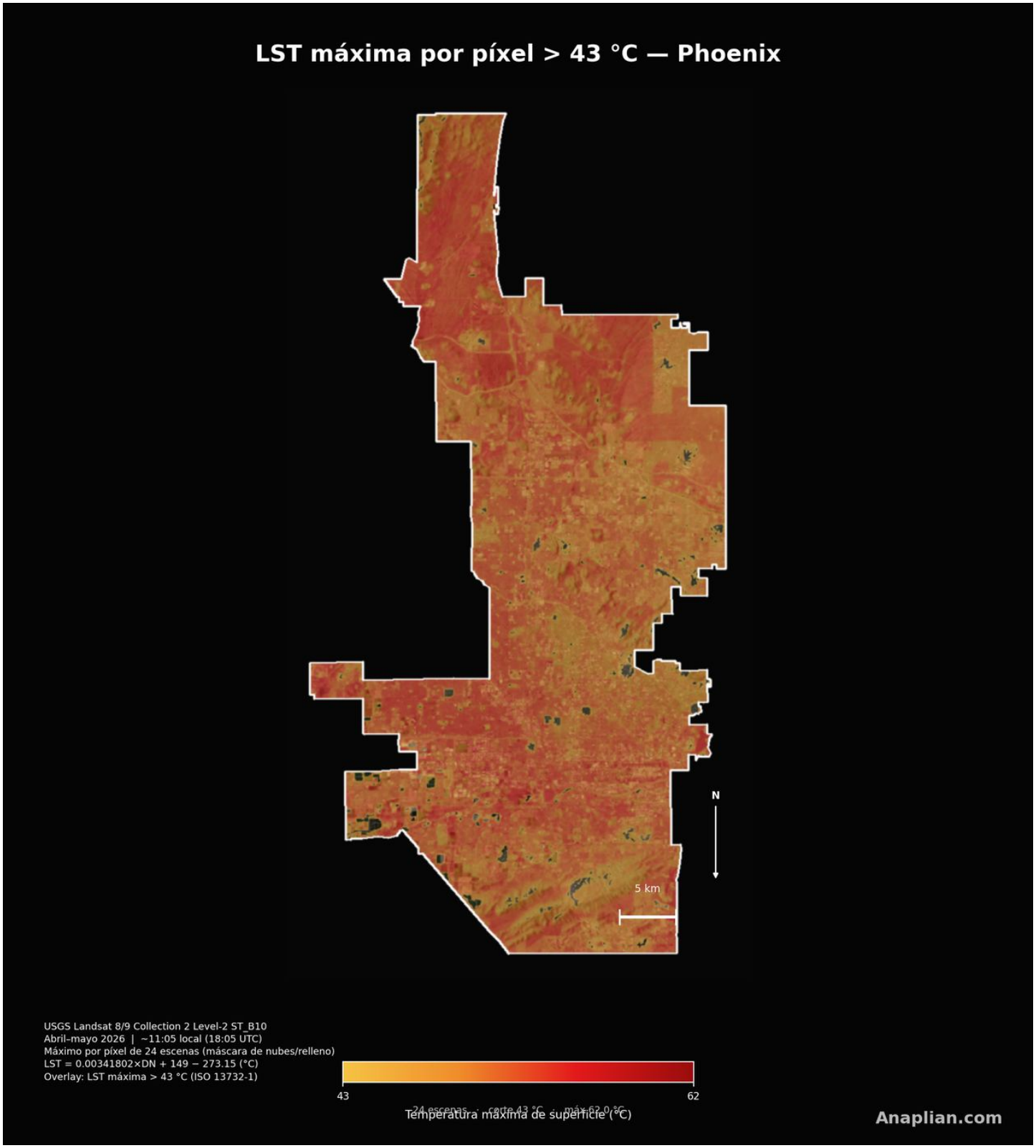
Highest-scoring political subdivisions in the index.



Index components of the highest-scoring subdivisions.

4. Per-pixel maximum LST > 43 °C

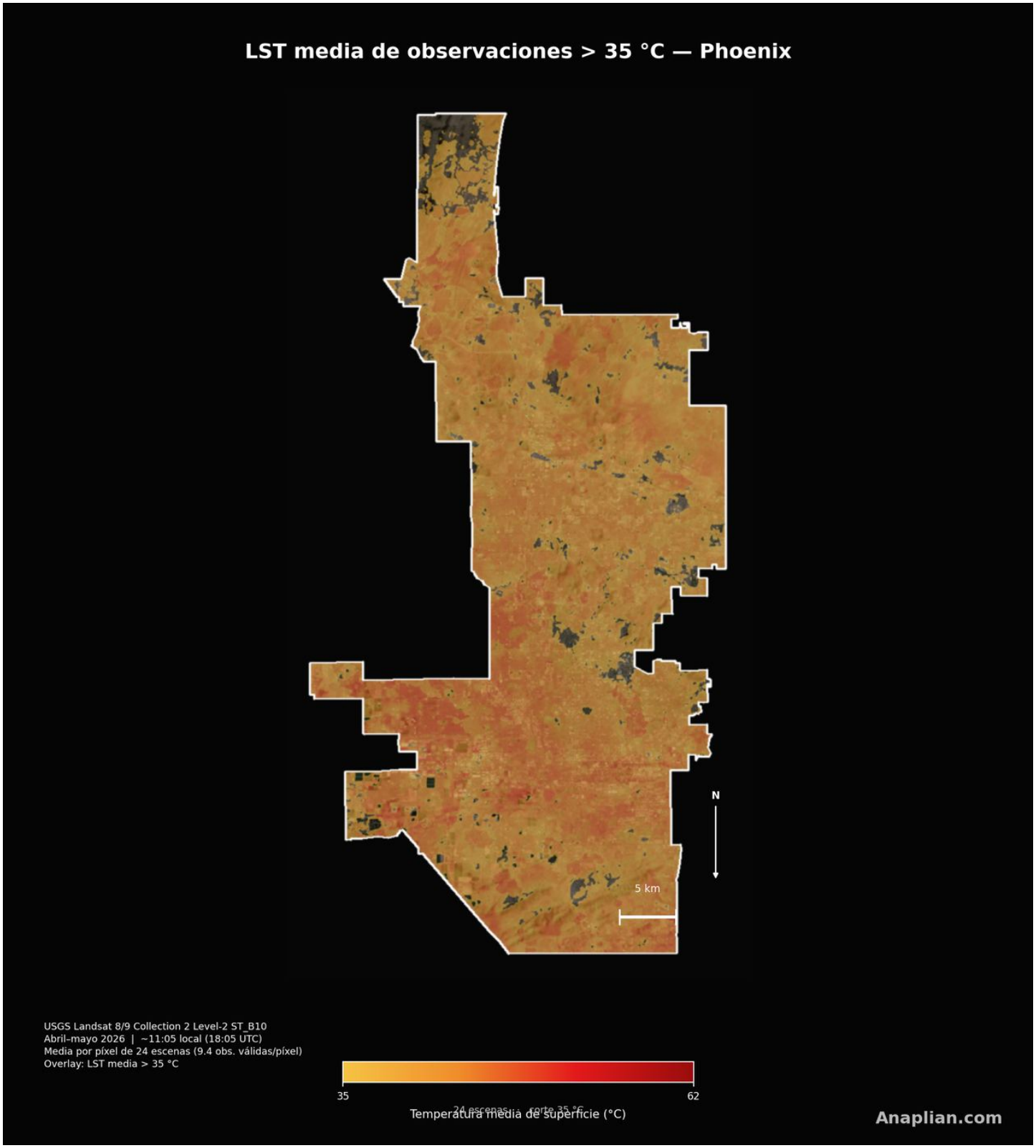
Maximum surface temperature. Landsat 8 and 9 Collection 2 Level 2, band 10 (ST_B10), 1 April–31 May 2026. 24 scenes; overpass ~11:05 local time (18:05 UTC). Per-pixel mask of fill, cloud, shadow and LST outside 0–80 °C. LST = 0.00341802×DN + 149 − 273.15. Maximum of valid observations (9.4 obs./pixel). Overlay and index: urban fraction > 43 °C.



24 scenes · ~11:05 local time (18:05 UTC) · ISO overlay 43 °C.

5. Mean LST of observations > 35 °C

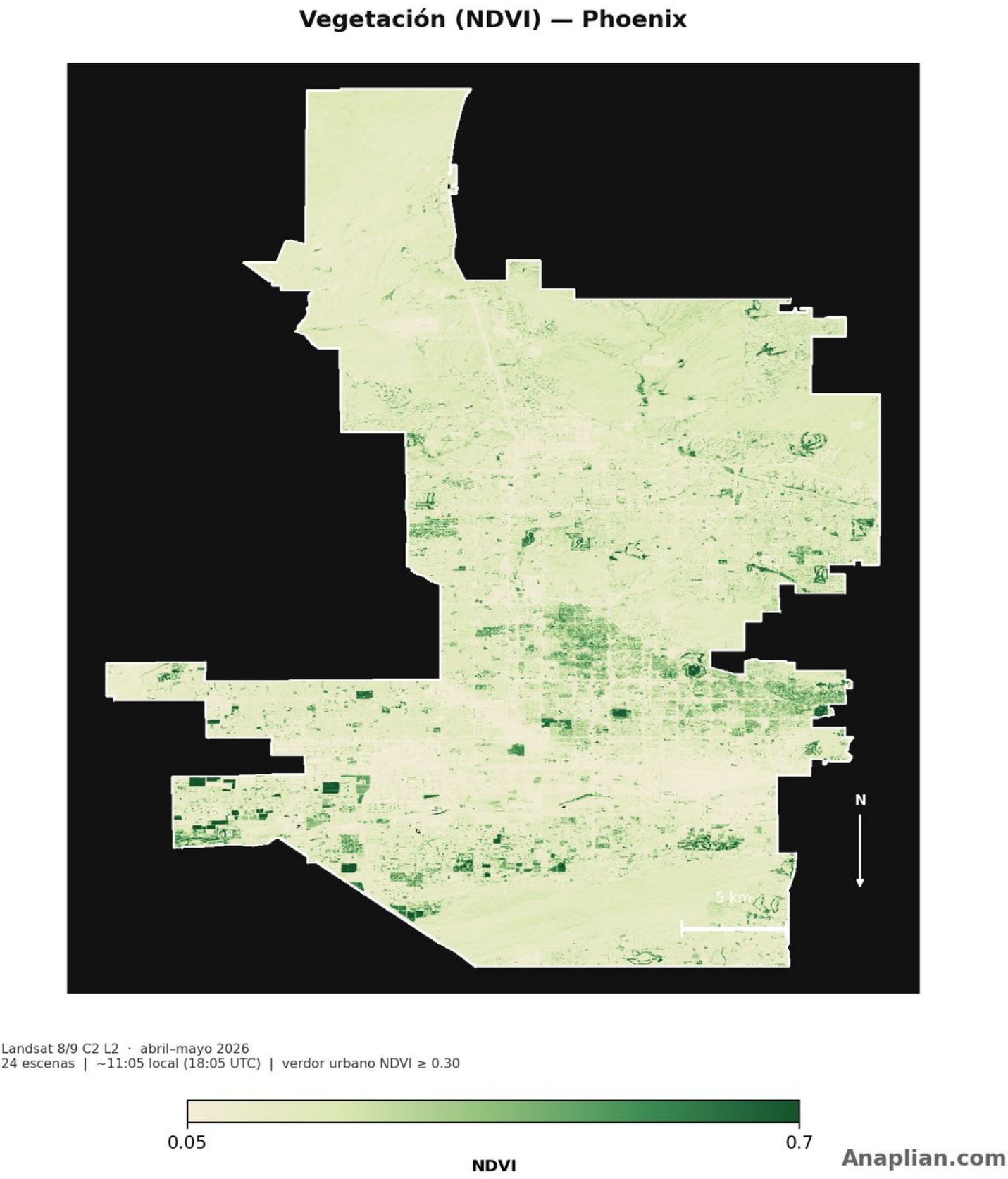
Mean surface temperature of observations. Same collection and mask. Mean of valid scenes. The map colours from 35 °C; the index uses the urban fraction > 43 °C.



Mean of valid observations. Index: urban fraction > 43 °C.

6. Vegetation (NDVI)

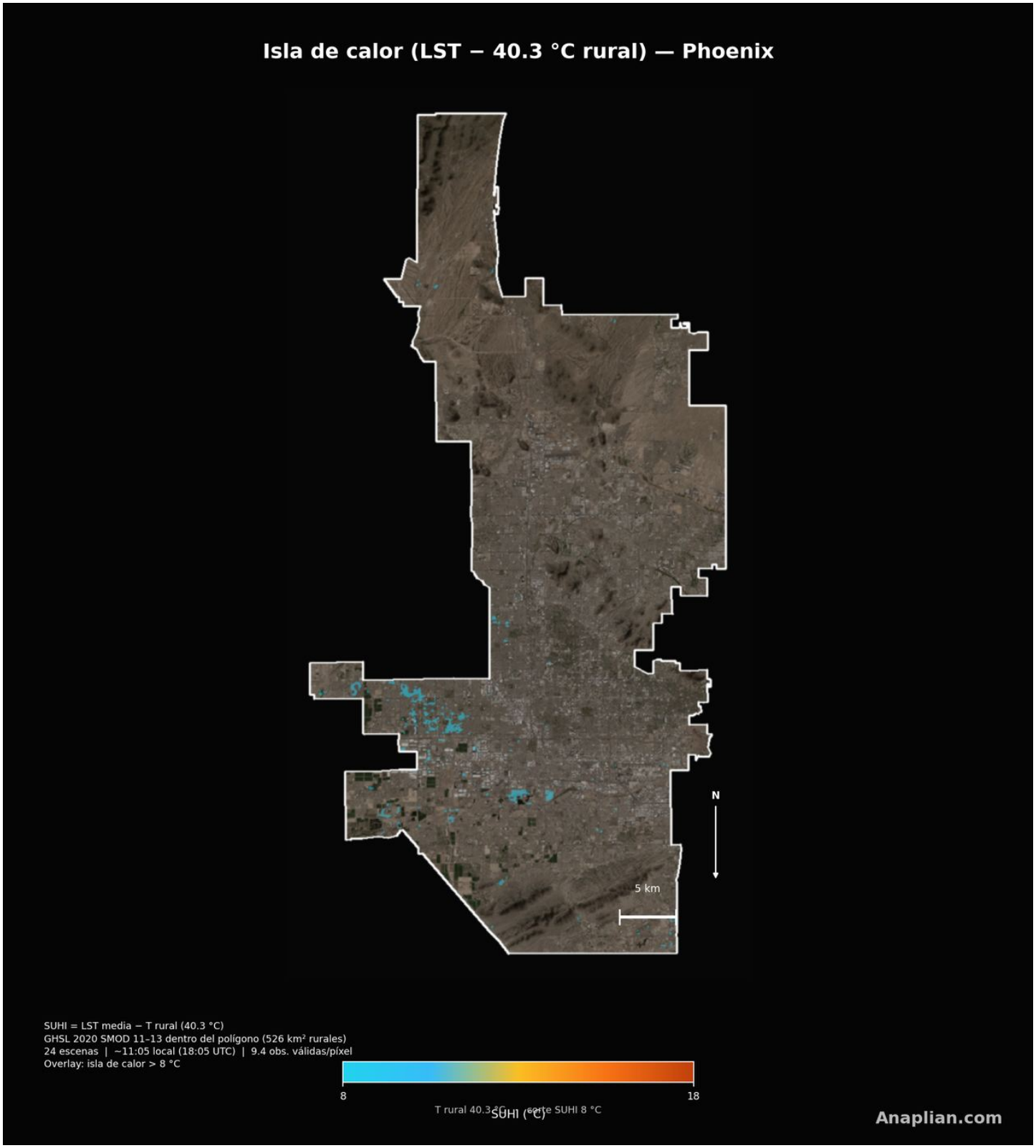
Vegetation. NDVI = (NIR − red)/(NIR + red), SR_B5 and SR_B4. Period mean (24 scenes, ~11:05 local time (18:05 UTC)). Urban greenness: NDVI ≥ 0.30 inside GHSL SMOD ≥ 21. It enters the index inverted on a fixed 0.05–0.55 scale.



Mean NDVI. Urban greenness ≥ 0.30.

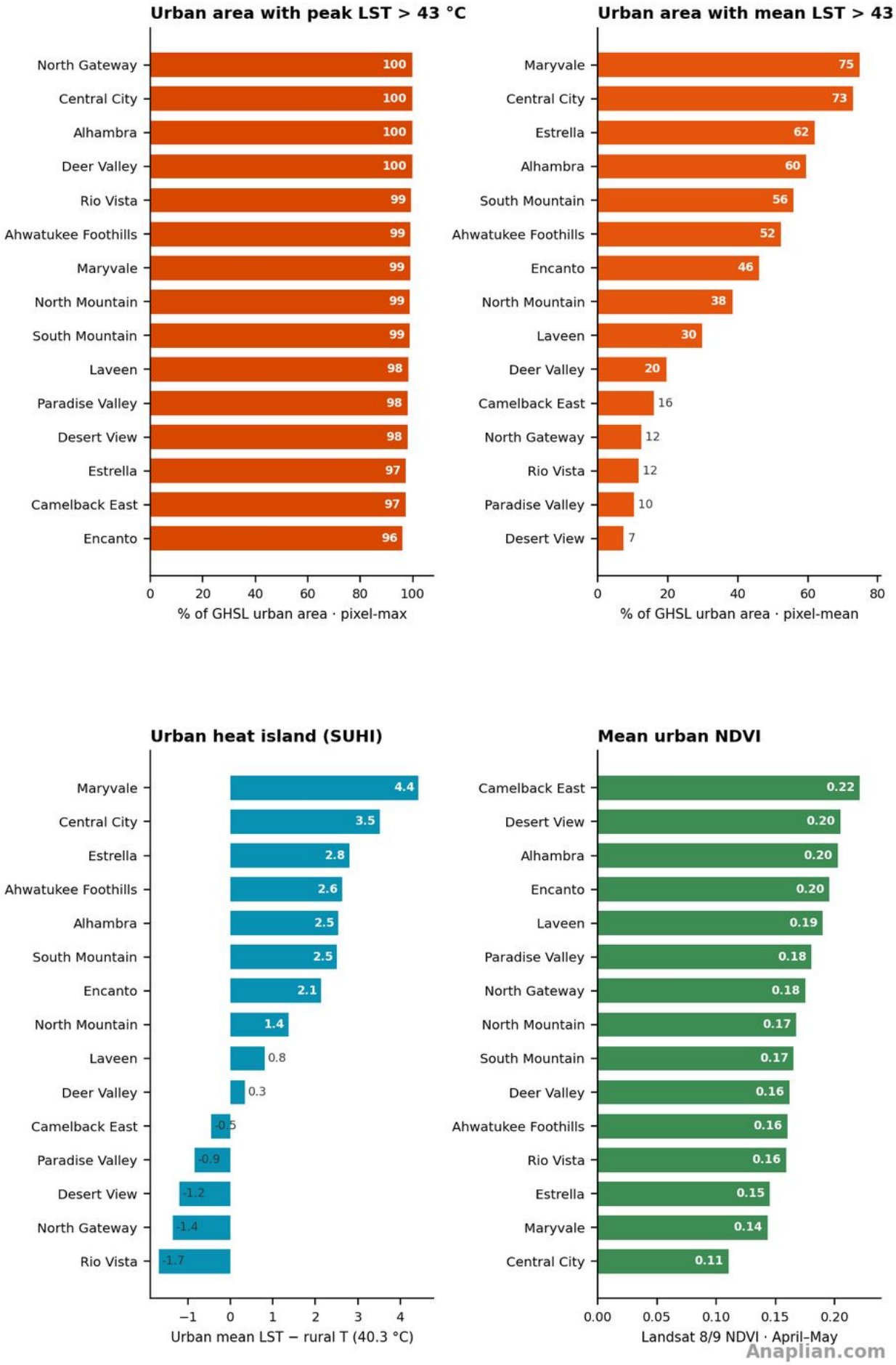
7. Heat island (pixel − rural)

Surface urban heat island. SUHI = pixel mean LST − rural T. T rural = 40.3 °C, GHSL 2020 SMOD 11–13 inside the polygon (526 km² rural). Overlay: SUHI > 8 °C. z_SUHI = min(max(urban SUHI, 0) / 15, 1).



SUHI = mean LST − 40.3 °C. Overlay > 8 °C.

8. Four sub-index elements



9. Table of urban villages

#	Urban vill	IVC	Max43%	Med43%	SUHI	NDVI
1	Maryvale	71.2	99	75	4.4	0.14
2	Central City	71.1	100	73	3.5	0.11
3	Estrella	64.8	97	62	2.8	0.15
4	South Mountain	62.1	99	56	2.5	0.17
5	Ahwatukee Foothills	61.8	99	52	2.6	0.16
6	Alhambra	61.4	100	60	2.5	0.20
7	Encanto	56.9	96	46	2.1	0.20
8	North Mountain	55.7	99	38	1.4	0.17
9	Laveen	51.4	98	30	0.8	0.19
10	Deer Valley	49.8	100	20	0.3	0.16
11	Rio Vista	47.3	99	12	-1.7	0.16
12	North Gateway	46.8	100	12	-1.4	0.18
13	Paradise Valley	45.6	98	10	-0.9	0.18
14	Camelback East	44.7	97	16	-0.5	0.22
15	Desert View	43.6	98	7	-1.2	0.20

10. Urban villages

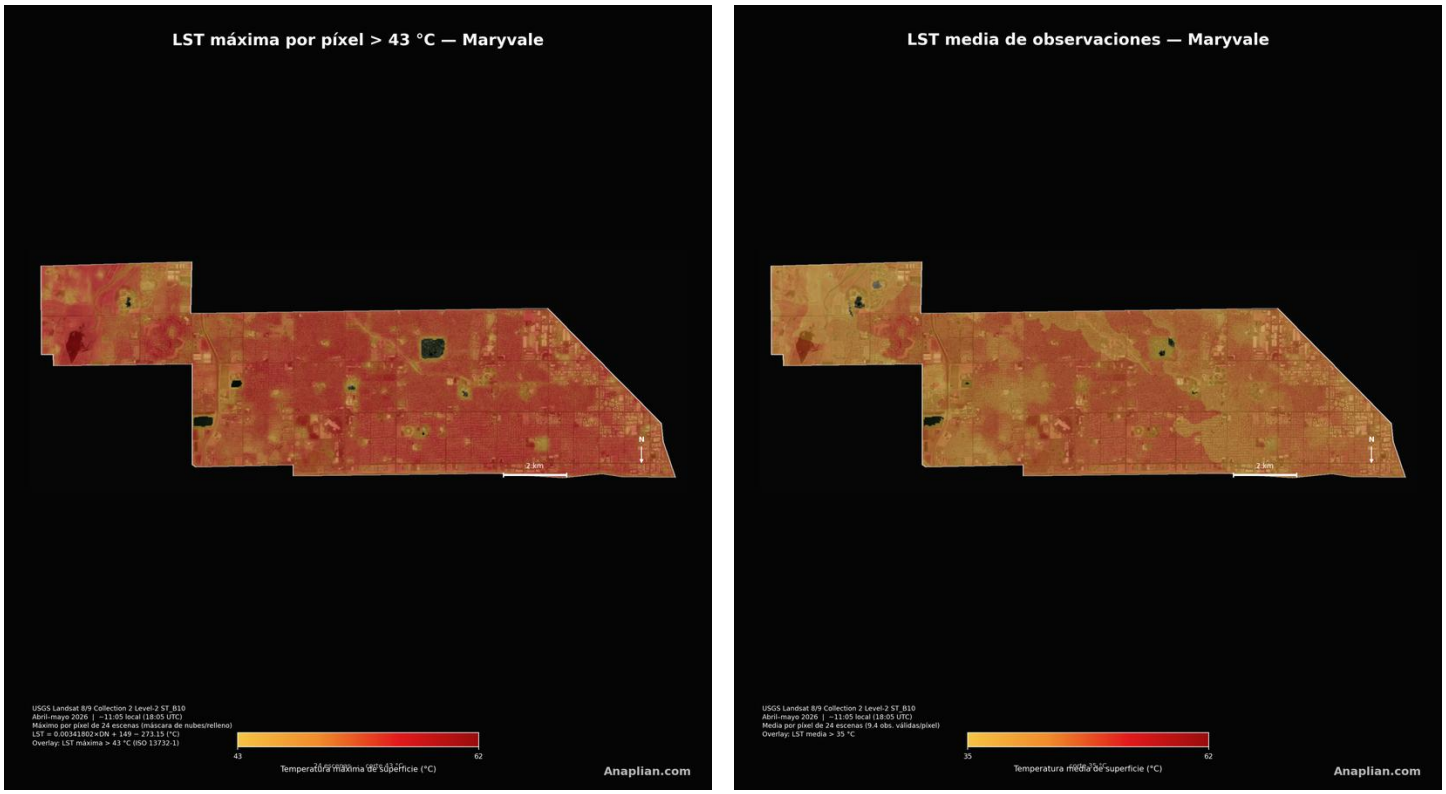
1. Maryvale · IVC 71.2

The most vulnerable urban village (IVC 71). Arterials, parking and low roofs on the west side; NDVI 0.14, 75 % of the urban mean > 43 °C. Slab on desert, not a CDMX-style differential.

LST max > 43 C: 99% urban (52.6 C). LST mean > 43 C: 75% urban (44.8 C). SUHI 4.4 C vs T rural 40.3 C. NDVI 0.14; 81.4 km2; 6.5 obs./pixel.

On the west side: six-lane arterials, strip-mall parking and low roofs. Almost no canopy; asphalt and slab concrete explain 75 % of the mean > 43 °C.

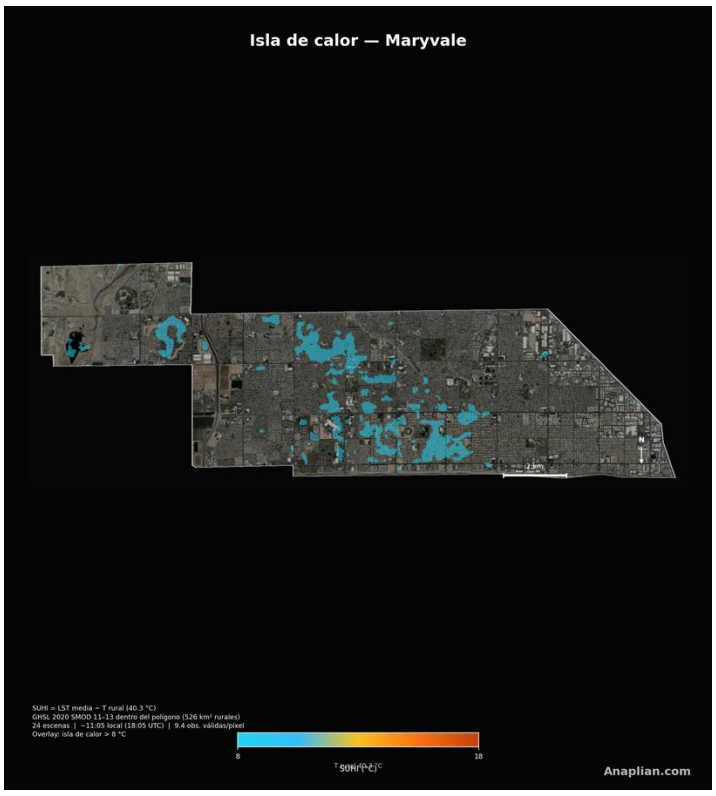
Peak LST > 43 C



Mean LST > 35 C



NDVI



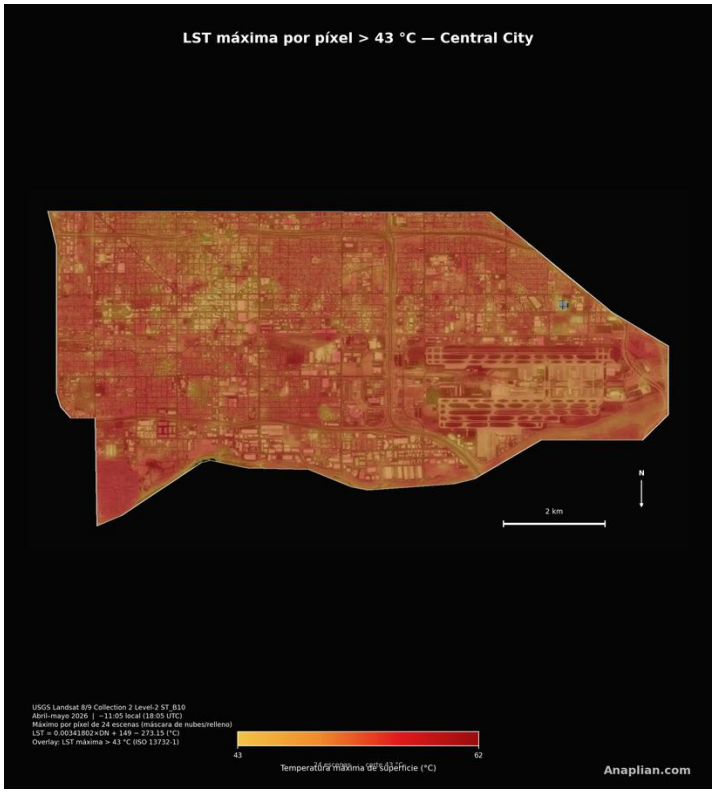
SUHI vs rural

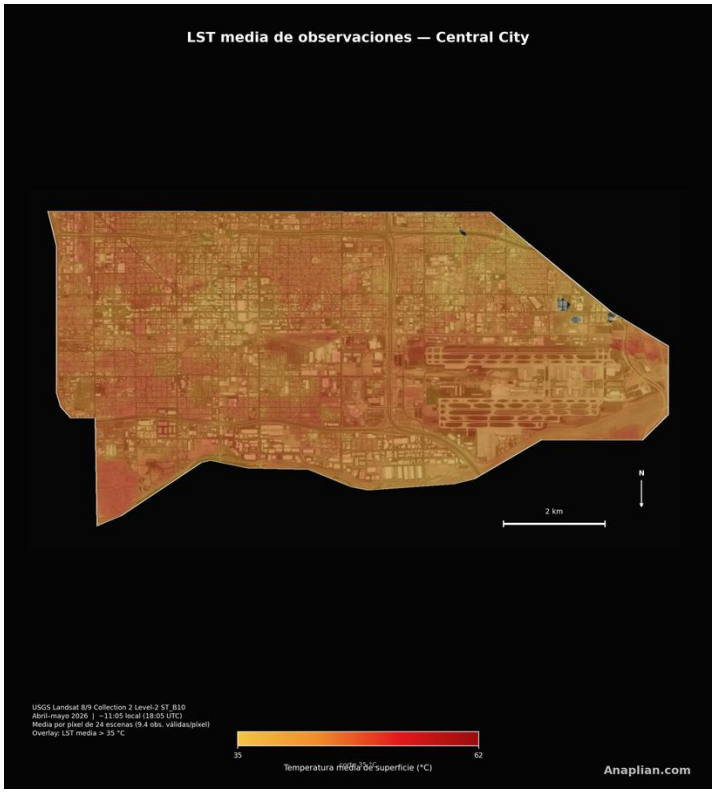
2. Central City · IVC 71.1

Core and warehouses (IVC 71). Phoenix’s lowest urban NDVI (0.11): parking, overpasses and the Sky Harbor airport edge. 100 % of pixel-max and 73 % of the mean > 43 °C.

LST max > 43 C: 100% urban (51.7 C). LST mean > 43 C: 73% urban (43.9 C). SUHI 3.5 C vs T rural 40.3 C. NDVI 0.11; 39.6 km2; 7.0 obs./pixel.

In the core: downtown, warehouses and the Sky Harbor edge. Runways, aprons and parking replace vegetation; NDVI 0.11 is the city’s floor.

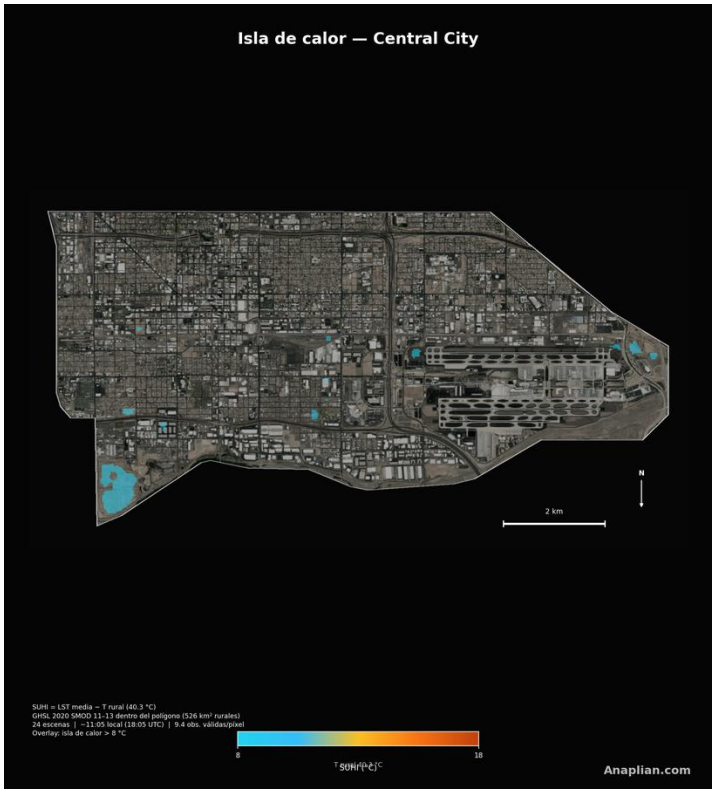




Peak LST > 43 C

Mean LST > 35 C





NDVI

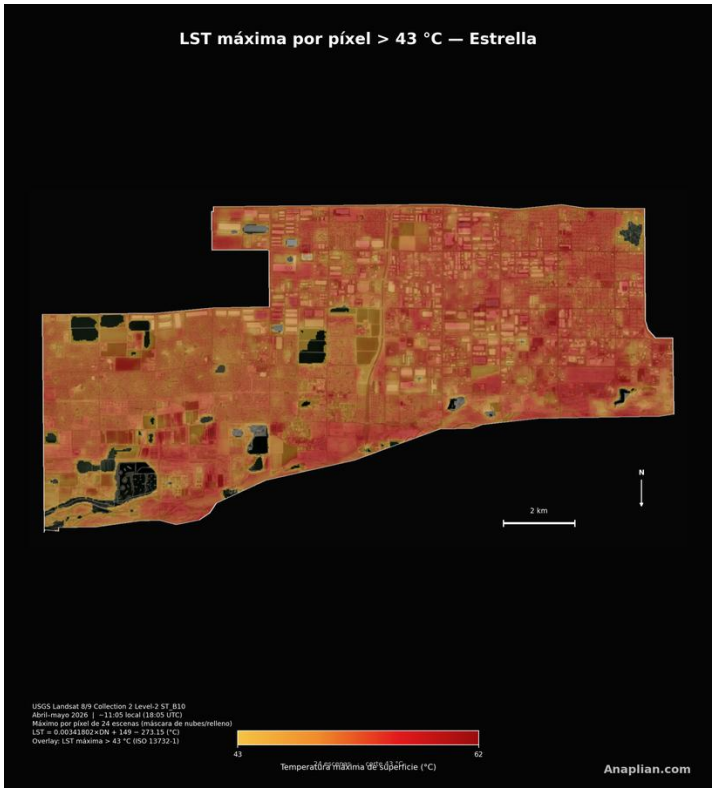
SUHI vs rural

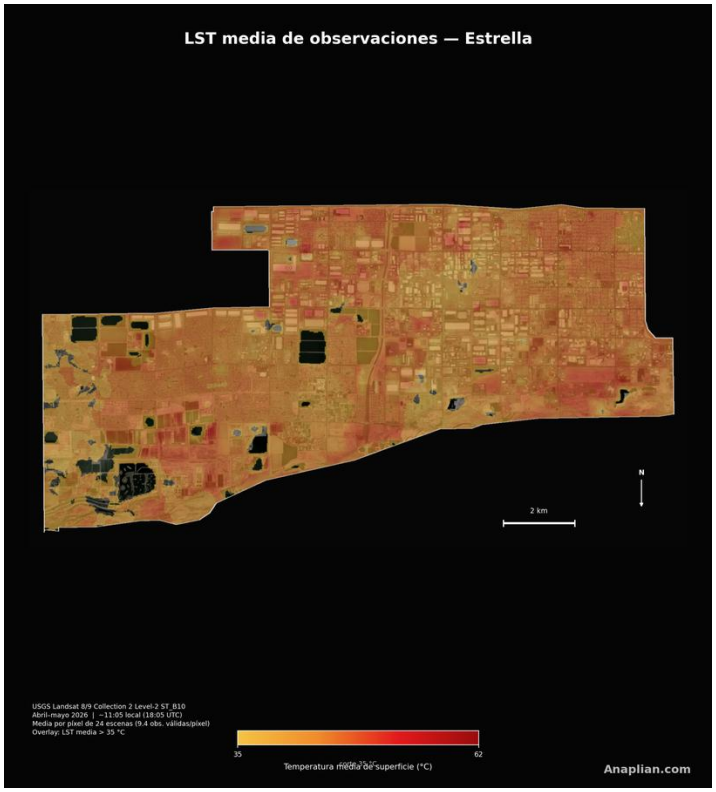
3. Estrella · IVC 64.8

Southwest expansion (IVC 65). New housing on desert: fresh asphalt, roofs and few trees (NDVI 0.15). 62 % of the urban mean already crosses 43 °C.

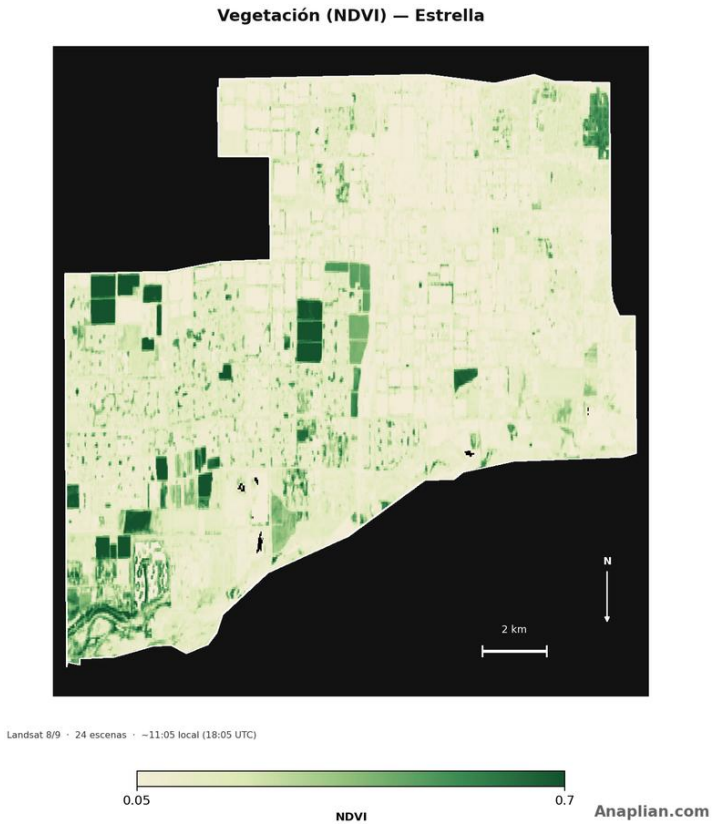
LST max > 43 C: 97% urban (51.2 C). LST mean > 43 C: 62% urban (43.2 C). SUHI 2.8 C vs T rural 40.3 C. NDVI 0.15; 76.1 km2; 6.4 obs./pixel.

In the southwest expansion: new streets, roofs and scraped desert. Fresh asphalt on bare soil pushes the mean to 62 % above 43 °C.



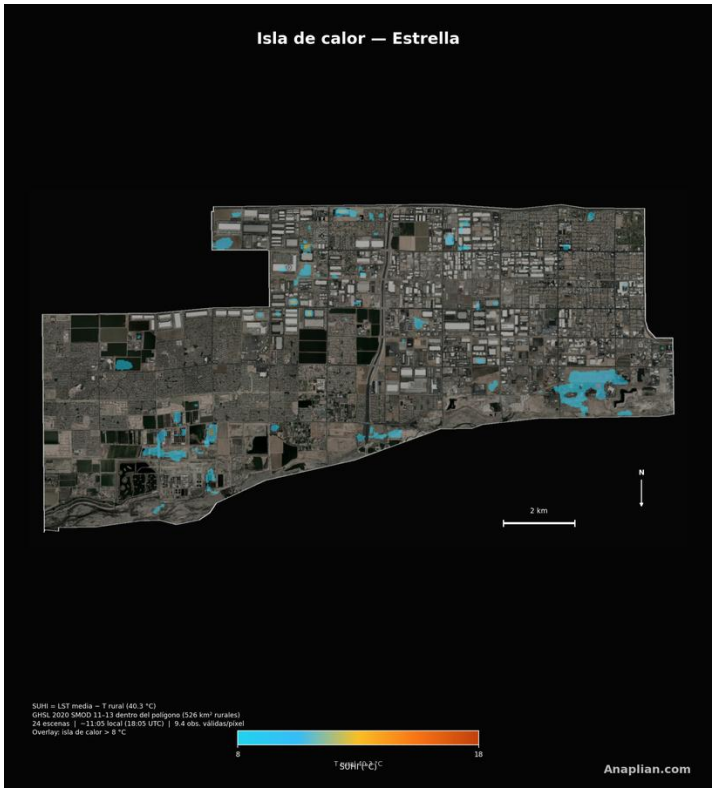


Peak LST > 43 C



NDVI

Mean LST > 35 C



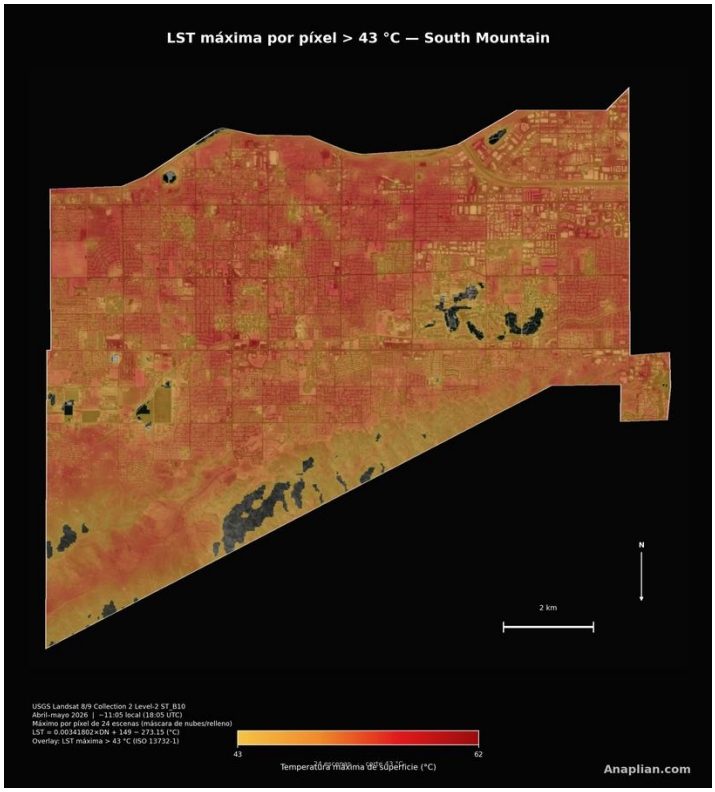
SUHI vs rural

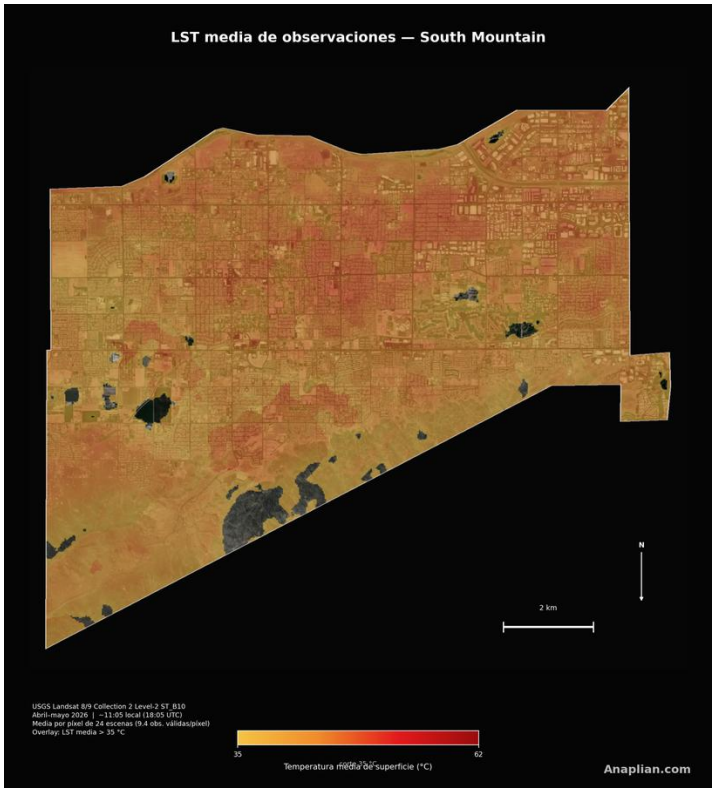
4. South Mountain · IVC 62.1

Preserve versus paved foothills (IVC 62). South Mountain Preserve is not in the urban index; GHSL ≥ 21 sits on the asphalted slopes, with 56 % of the mean > 43 °C.

LST max > 43 C: 99% urban (50.8 C). LST mean > 43 C: 56% urban (42.9 C). SUHI 2.5 C vs T rural 40.3 C. NDVI 0.17; 77.5 km2; 7.7 obs./pixel.

On the foothills: the preserve is green; the index looks at hillside GHSL — houses, arterial and parking — not the mountain.

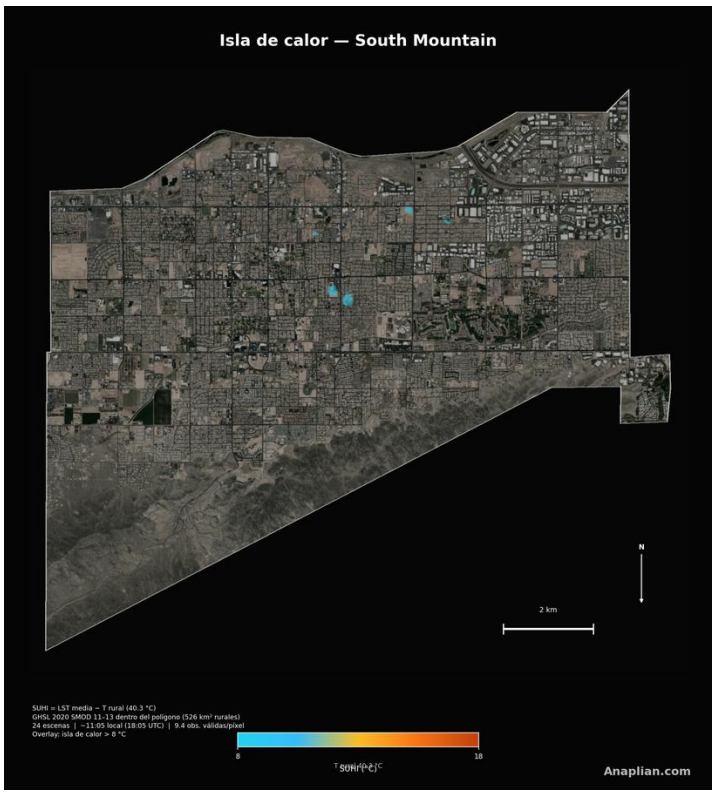




Peak LST > 43 C

Mean LST > 35 C





NDVI

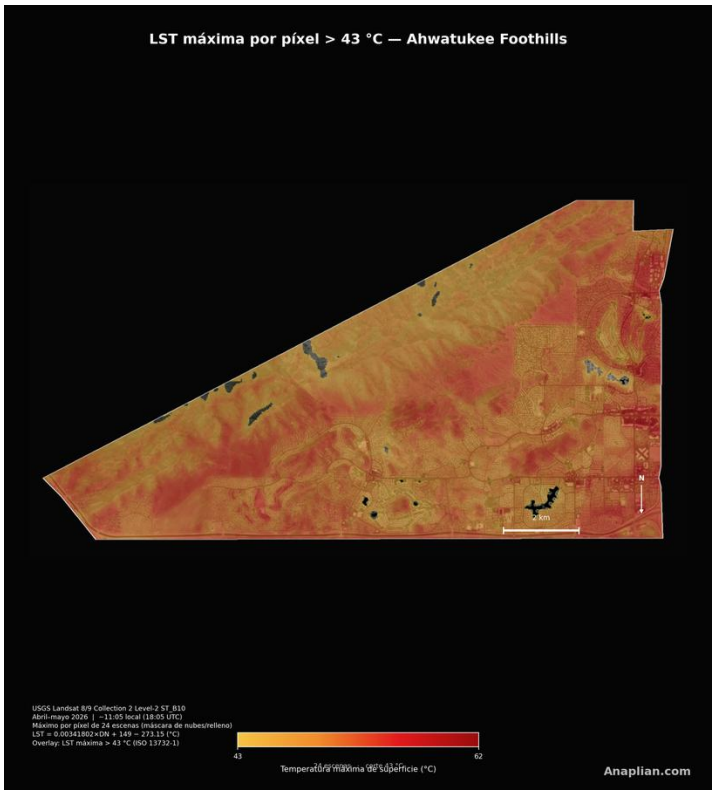
SUHI vs rural

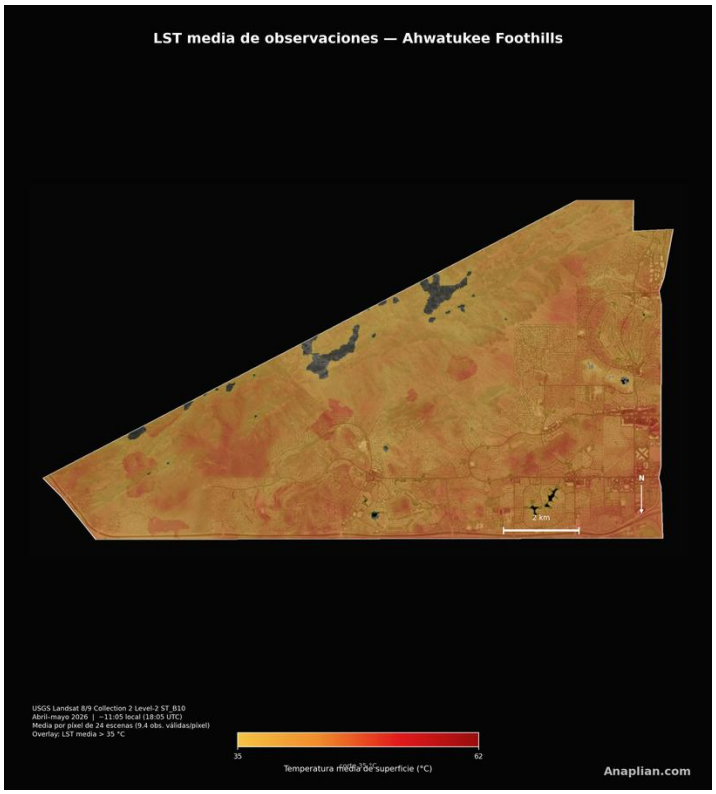
5. Ahwatukee Foothills · IVC 61.8

Master-planned foothills (IVC 62). Tile roofs, arterials and desert lots; 99 % of pixel-max > 43 °C and the mean still at 52 %.

LST max > 43 C: 99% urban (50.5 C). LST mean > 43 C: 52% urban (43.0 C). SUHI 2.6 C vs T rural 40.3 C. NDVI 0.16; 60.5 km2; 9.8 obs./pixel.

In the subdivision: tile, arterial and desert lot. Little street shade; roof-top maxima cross 43 °C on 99 % of urban land.

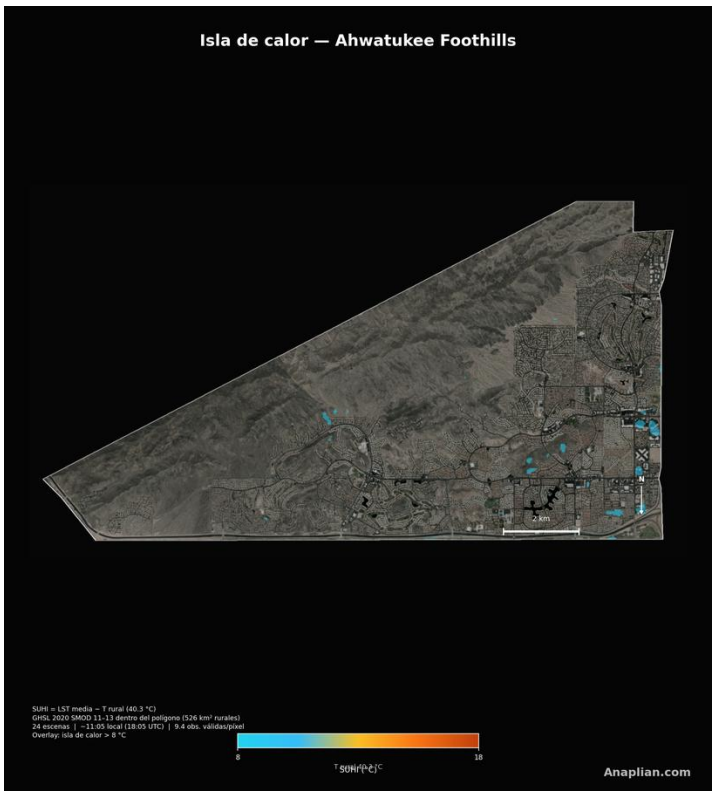




Peak LST > 43 C

Mean LST > 35 C





NDVI

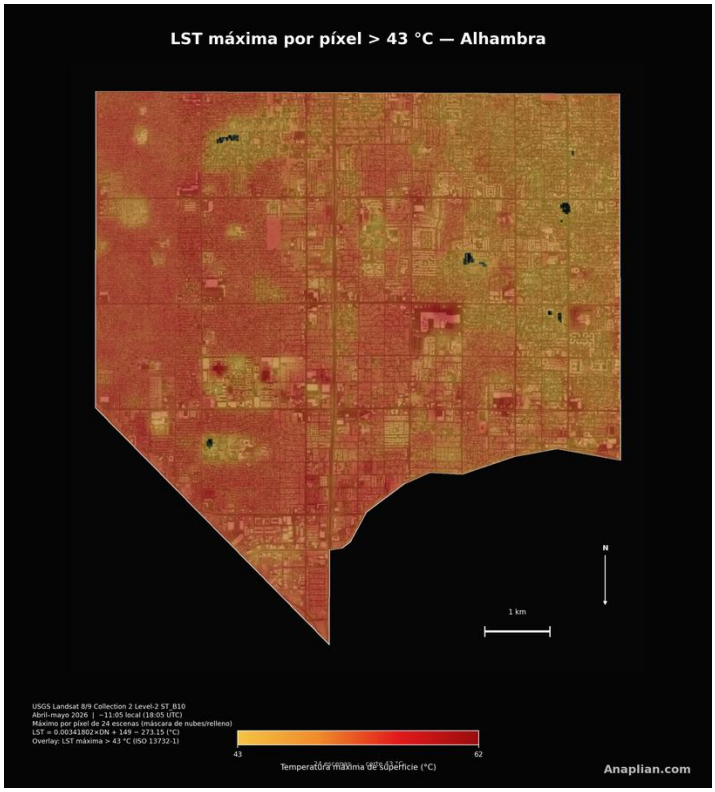
SUHI vs rural

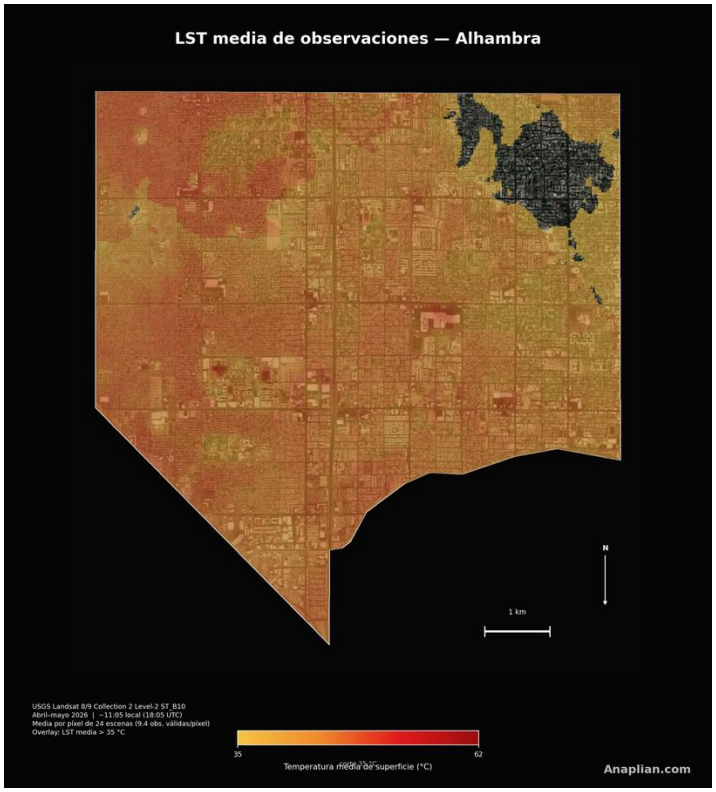
6. Alhambra · IVC 61.4

I-17 corridor (IVC 61). Commercial pads, asphalt and 60 % of the urban mean > 43 °C. A little more green than Maryvale (NDVI 0.20), not enough to cut the peak.

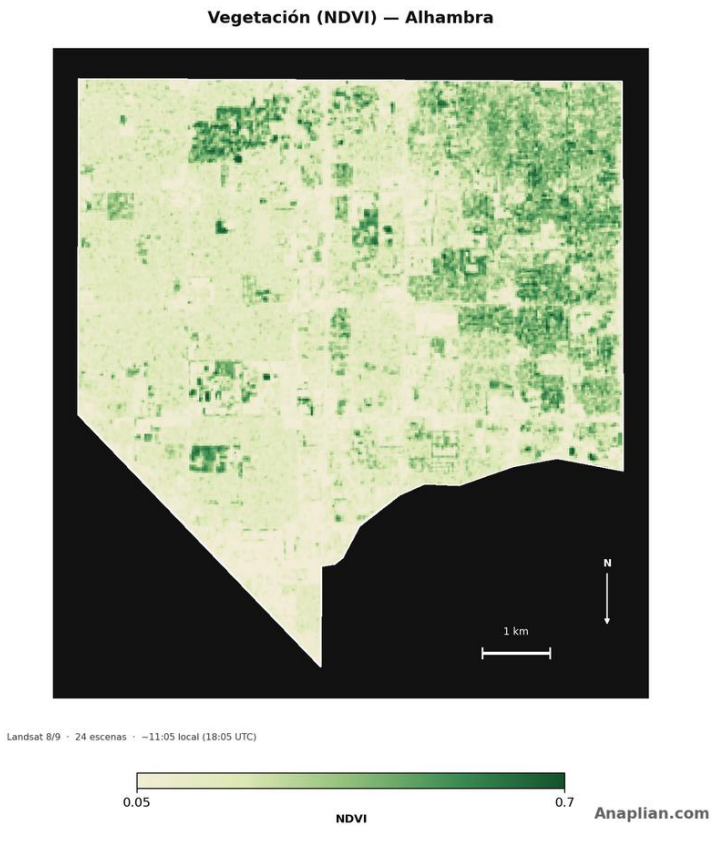
LST max > 43 C: 100% urban (50.7 C). LST mean > 43 C: 60% urban (42.9 C). SUHI 2.5 C vs T rural 40.3 C. NDVI 0.20; 49.8 km2; 6.9 obs./pixel.

Along I-17: commercial pads, stations and access asphalt. Fragmented canopy does not cut the 60 % mean above 43 °C.



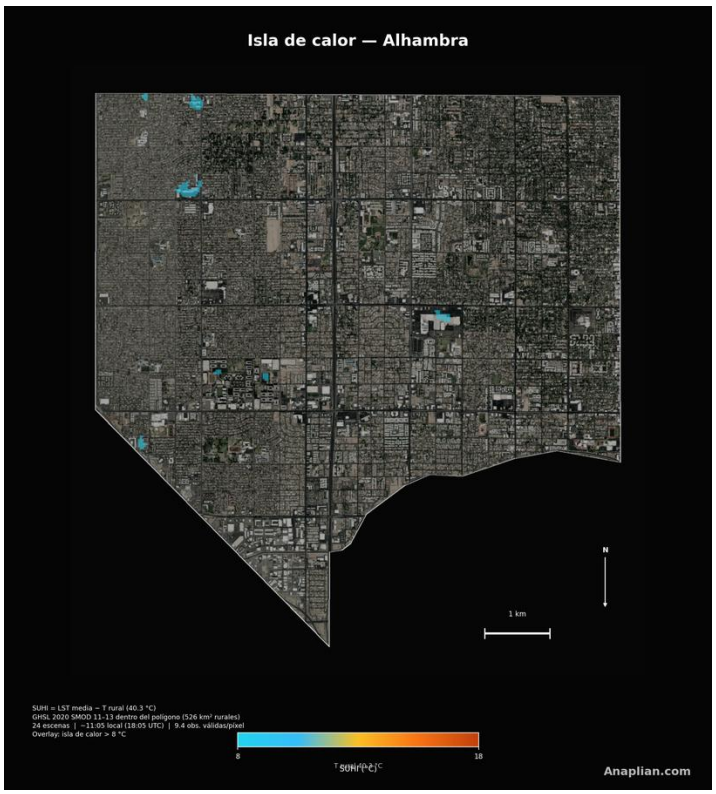


Peak LST > 43 C



NDVI

Mean LST > 35 C



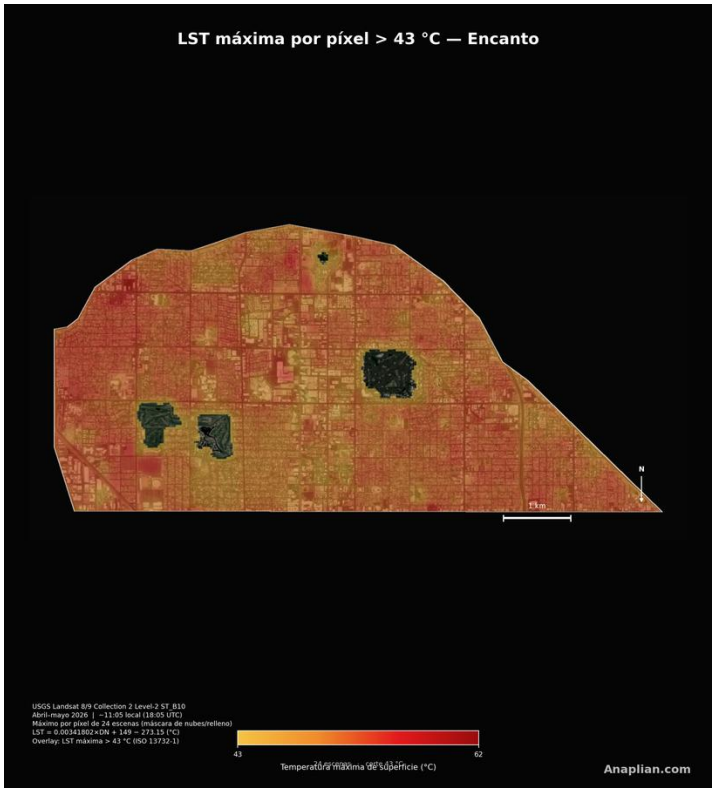
SUHI vs rural

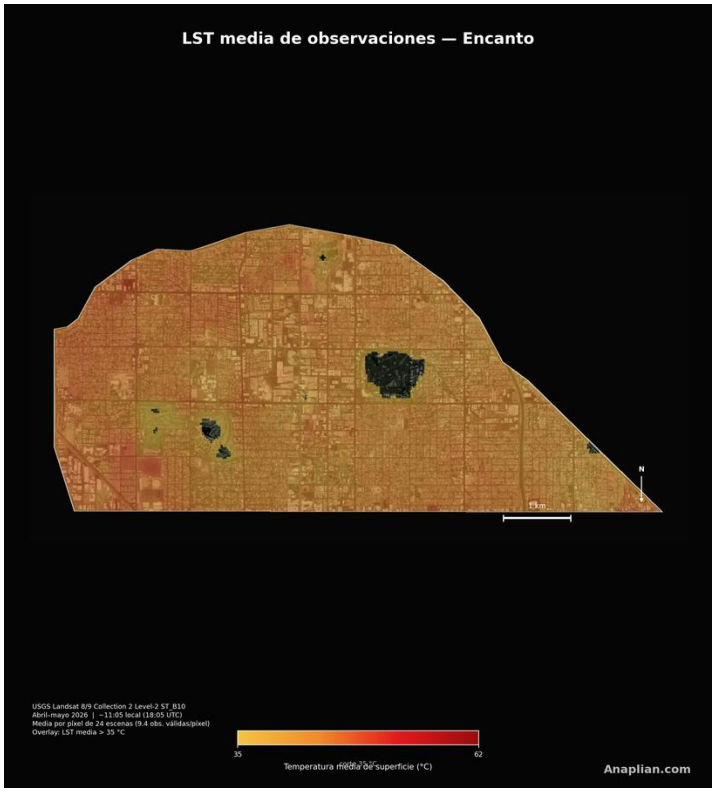
7. Encanto · IVC 56.9

Older grid (IVC 57). More trees than the west side; 46 % of the mean > 43 °C. Pixel-max is still 96 %: roof peak in a neighbourhood with some canopy.

LST max > 43 C: 96% urban (50.3 C). LST mean > 43 C: 46% urban (42.5 C). SUHI 2.1 C vs T rural 40.3 C. NDVI 0.20; 27.1 km2; 7.0 obs./pixel.

On the older grid: more trees than Maryvale, still 96 % of pixel-max > 43 °C. Canopy lowers the mean, not the roof peak.



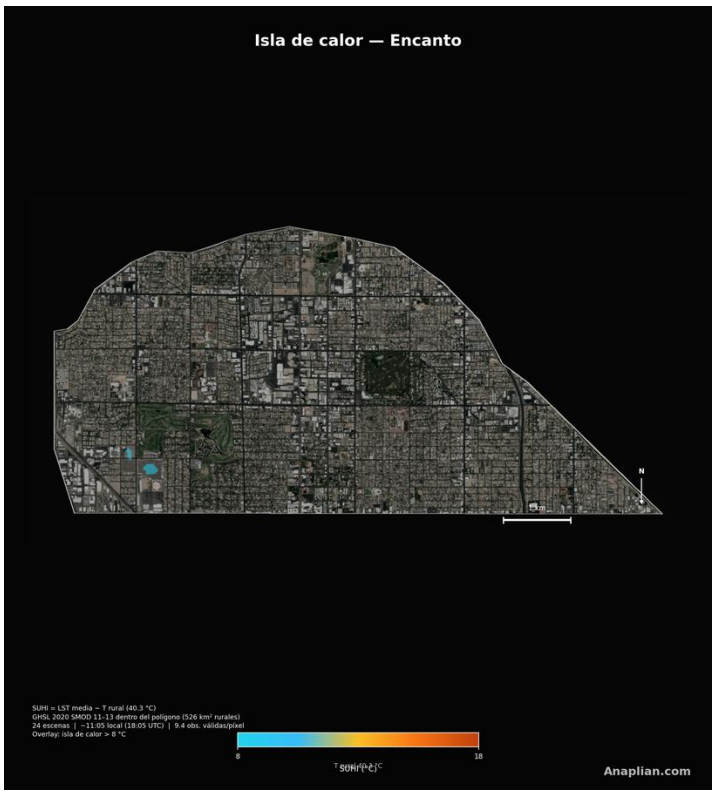


Peak LST > 43 C



NDVI

Mean LST > 35 C



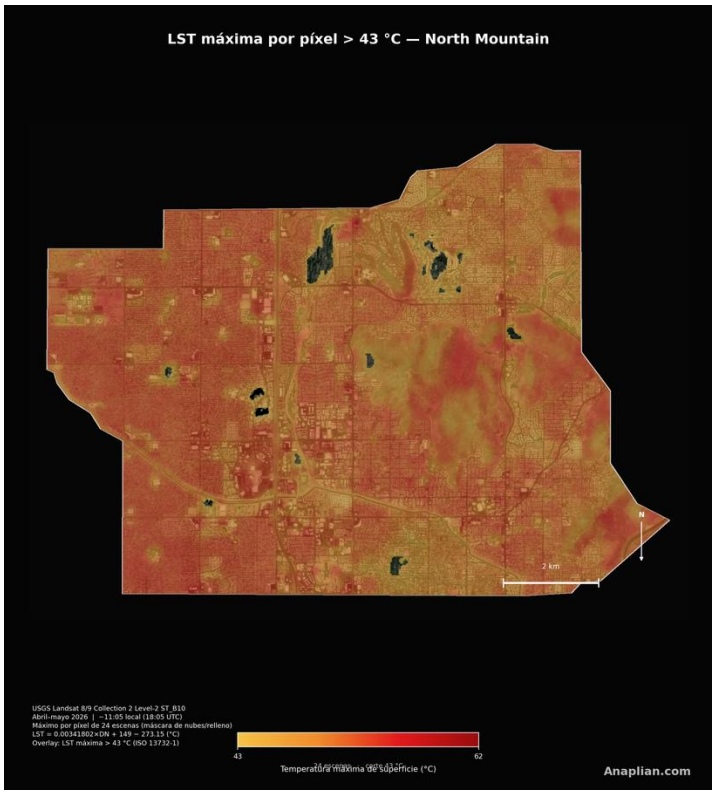
SUHI vs rural

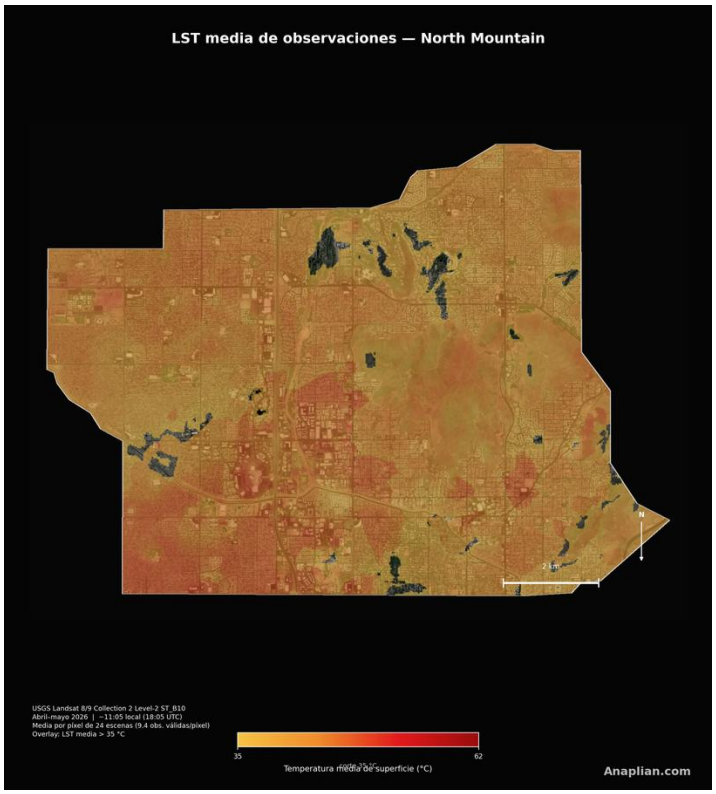
8. North Mountain · IVC 55.7

Mixed north (IVC 56). 99 % of pixel-max > 43 °C, but the mean drops to 38 %: less all-day load than Maryvale, the same midday slab.

LST max > 43 C: 99% urban (50.4 C). LST mean > 43 C: 38% urban (41.7 C). SUHI 1.4 C vs T rural 40.3 C. NDVI 0.17; 89.0 km2; 6.8 obs./pixel.

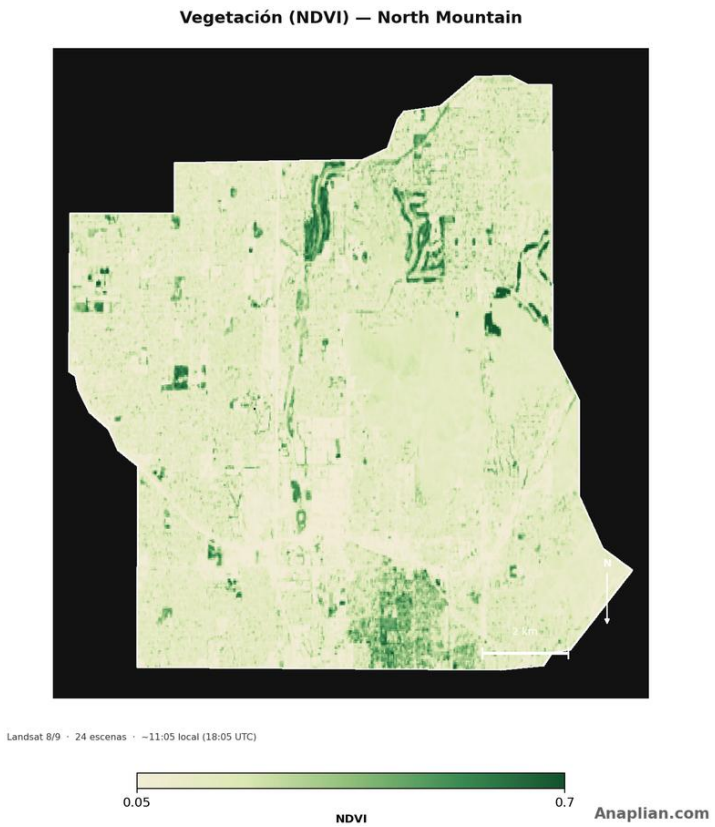
In the mixed north: arterial and roof; the mean (38 %) is no longer Maryvale's all-day load, the maximum still saturates.

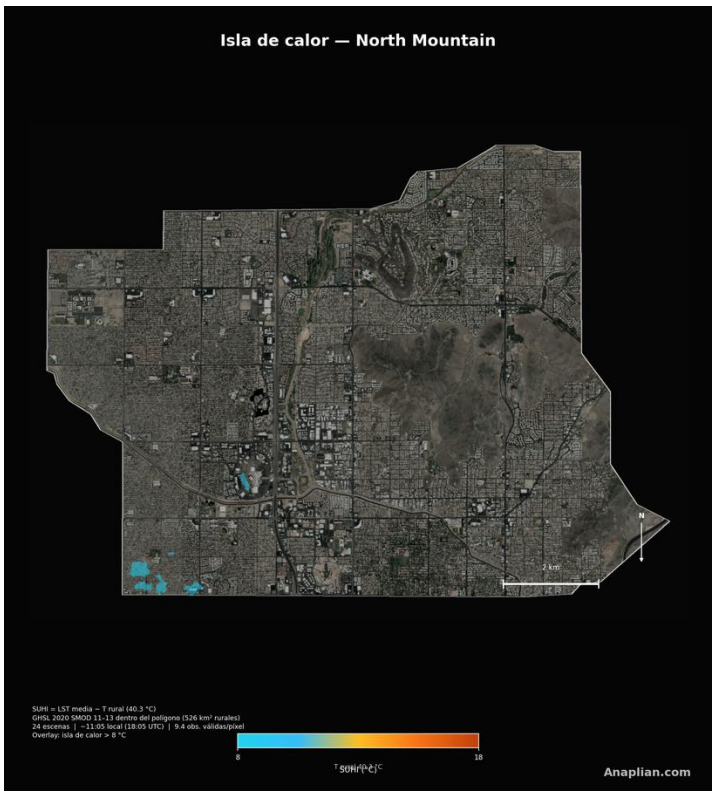




Peak LST > 43 C

Mean LST > 35 C





NDVI

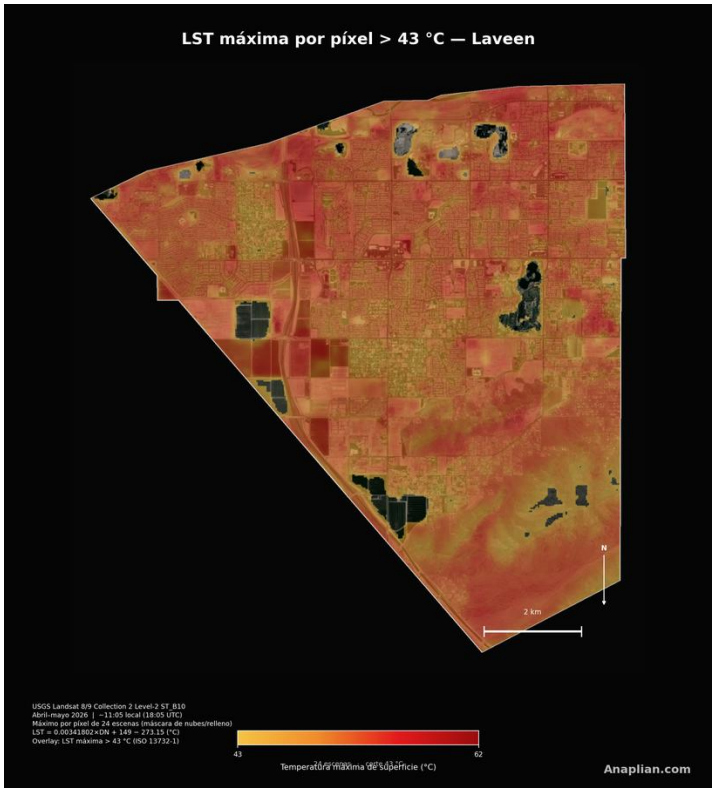
SUHI vs rural

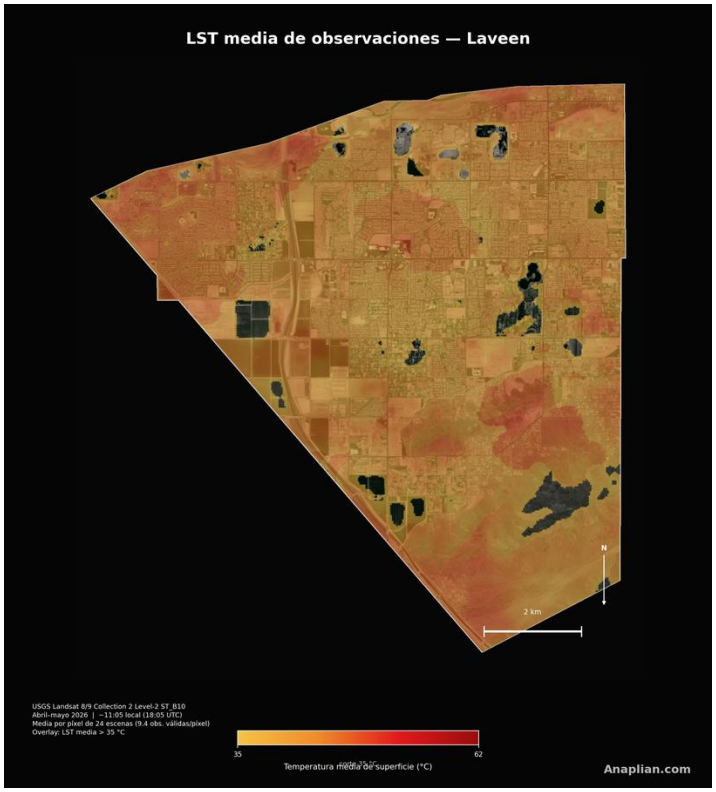
9. Laveen · IVC 51.4

Agricultural conversion (IVC 51). Fields turning into street and roof; 30 % of the urban mean > 43 °C. The index weights new GHSL land, not the remaining plot.

LST max > 43 C: 98% urban (50.7 C). LST mean > 43 C: 30% urban (41.2 C). SUHI 0.8 C vs T rural 40.3 C. NDVI 0.19; 39.4 km2; 6.8 obs./pixel.

On the conversion: agricultural plot next to a new street. The index takes the recent asphalt, not the remaining field.

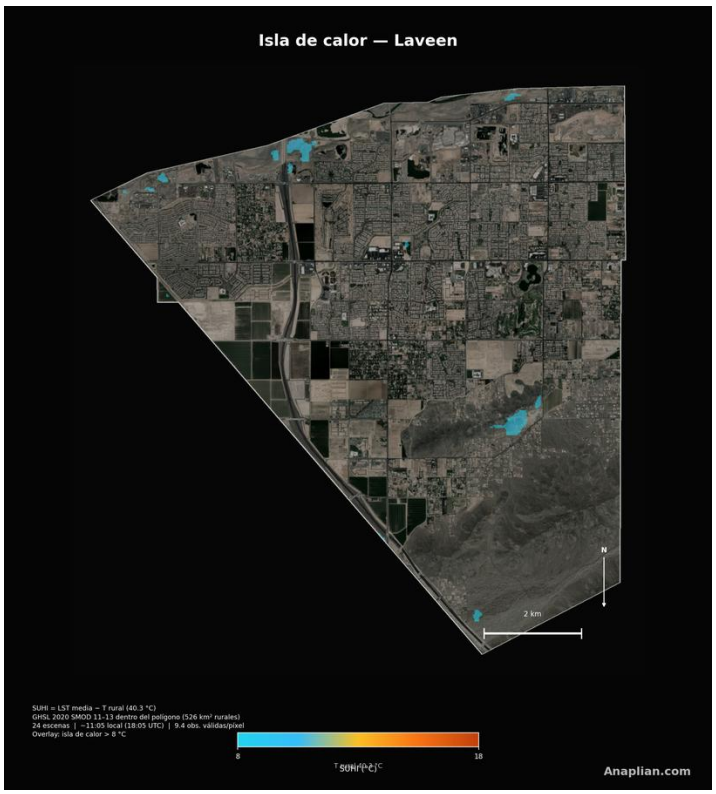




Peak LST > 43 C

Mean LST > 35 C





NDVI

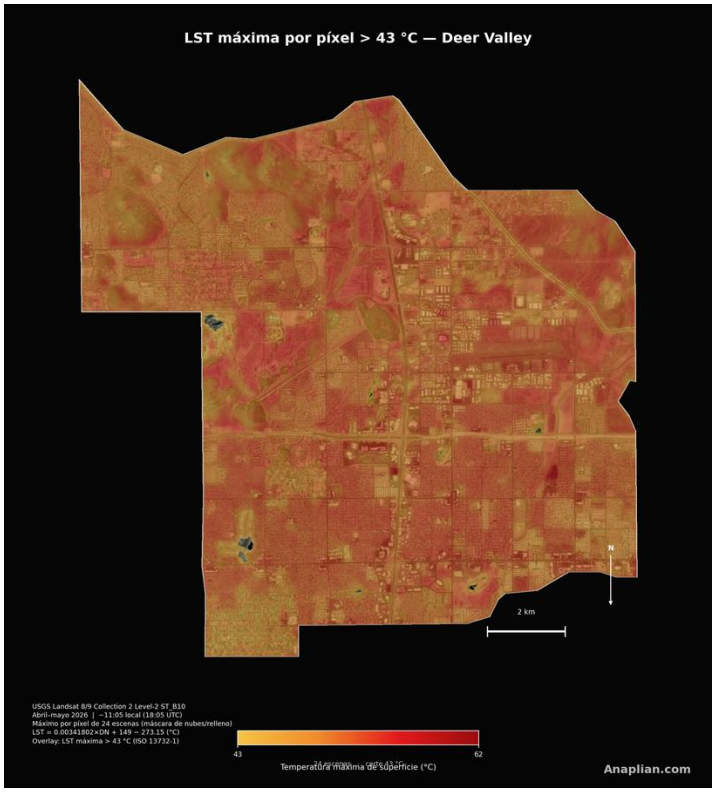
SUHI vs rural

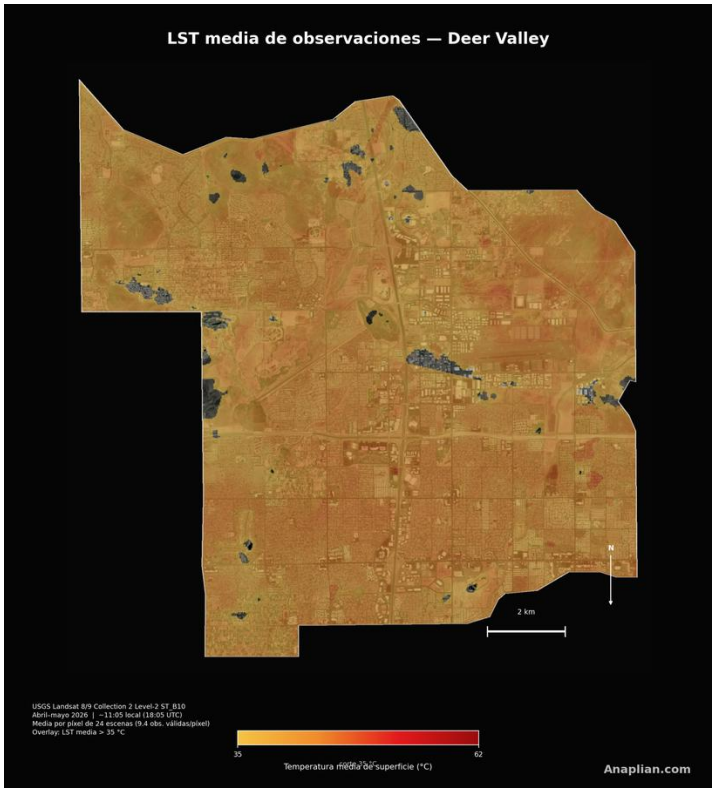
10. Deer Valley · IVC 49.8

Northern expansion (IVC 50). Almost all of pixel-max > 43 °C; the mean only 20 %. Roof peak in a more open urban village, by the northern airport.

LST max > 43 C: 100% urban (50.6 C). LST mean > 43 C: 20% urban (40.7 C). SUHI 0.3 C vs T rural 40.3 C. NDVI 0.16; 107.9 km2; 9.4 obs./pixel.

In the north: airport and expansion. Much roof-top max, little continuous mean: the land is not yet an all-day slab.





Peak LST > 43 C



NDVI

Mean LST > 35 C



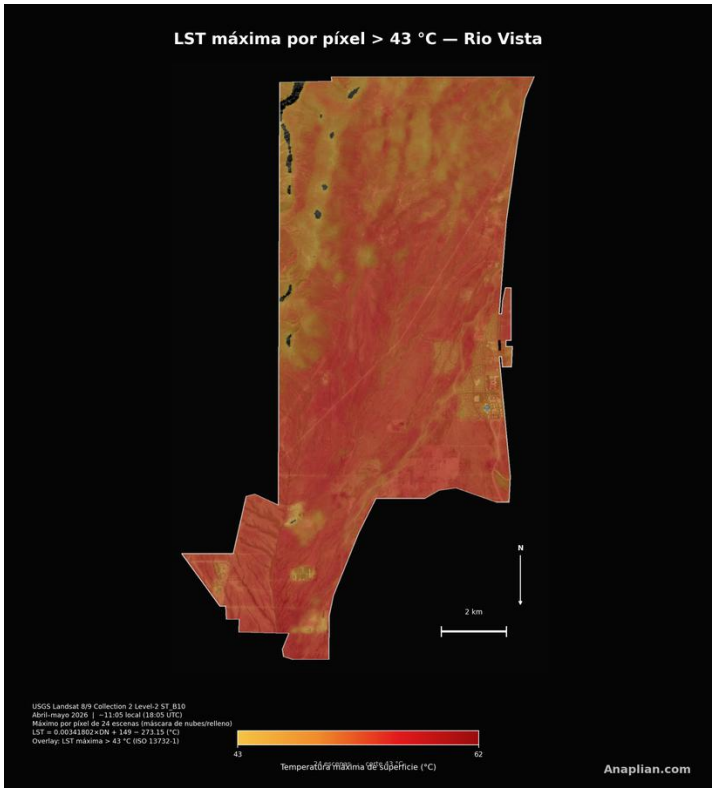
SUHI vs rural

11. Rio Vista · IVC 47.3

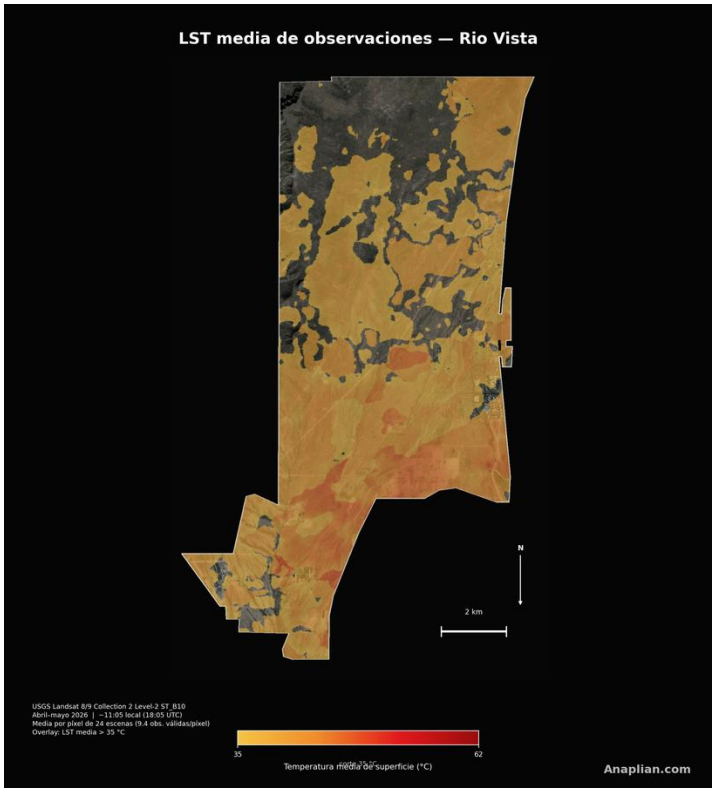
Canal and desert (IVC 47). Negative SUHI: rural land in the polygon already heats. 12 % of the mean > 43 °C; pixel-max still saturates.

LST max > 43 C: 99% urban (51.2 C). LST mean > 43 C: 12% urban (38.7 C). SUHI -1.7 C vs T rural 40.3 C. NDVI 0.16; 5.0 km2; 13.4 obs./pixel.

By the canal: agricultural water and roof. Rural land in the polygon heats; negative SUHI says the reference is not cool.



Peak LST > 43 C



Mean LST > 35 C



NDVI



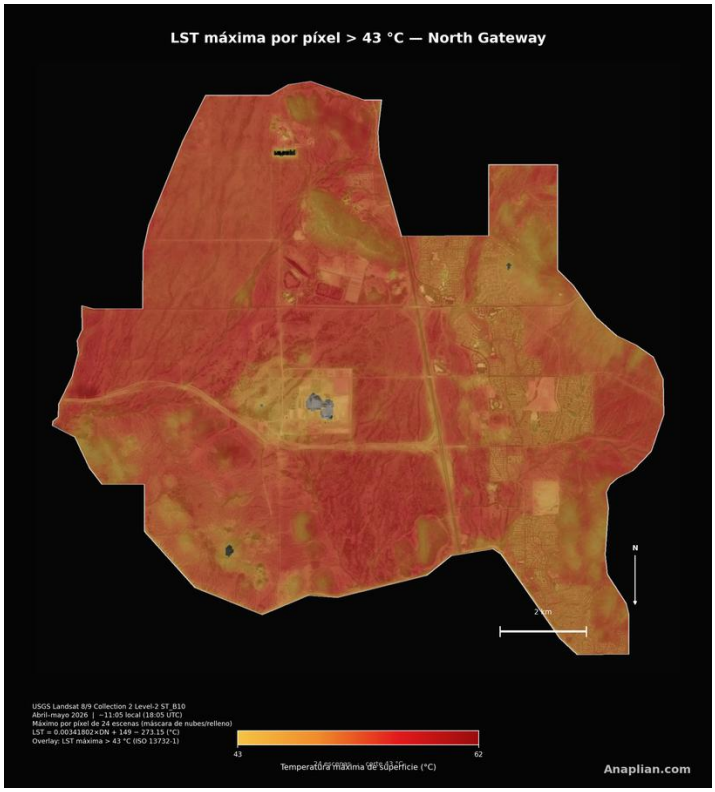
SUHI vs rural

12. North Gateway · IVC 46.8

Planned northern desert (IVC 47). SUHI −1.4 °C; 12 % of the mean > 43 °C. Phoenix's contrast: less continuous slab, the same April roof in the sun.

LST max > 43 C: 100% urban (50.5 C). LST mean > 43 C: 12% urban (39.0 C). SUHI -1.4 C vs T rural 40.3 C. NDVI 0.18; 19.8 km2; 13.5 obs./pixel.

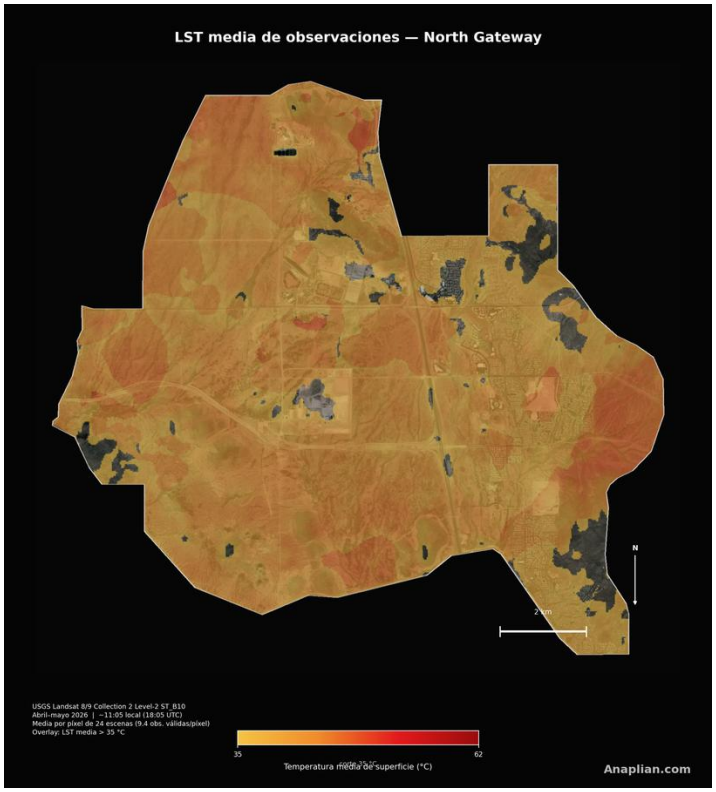
On the planned desert: low density, less slab. 12 % of the mean > 43 °C is the contrast with Maryvale, not a forest.



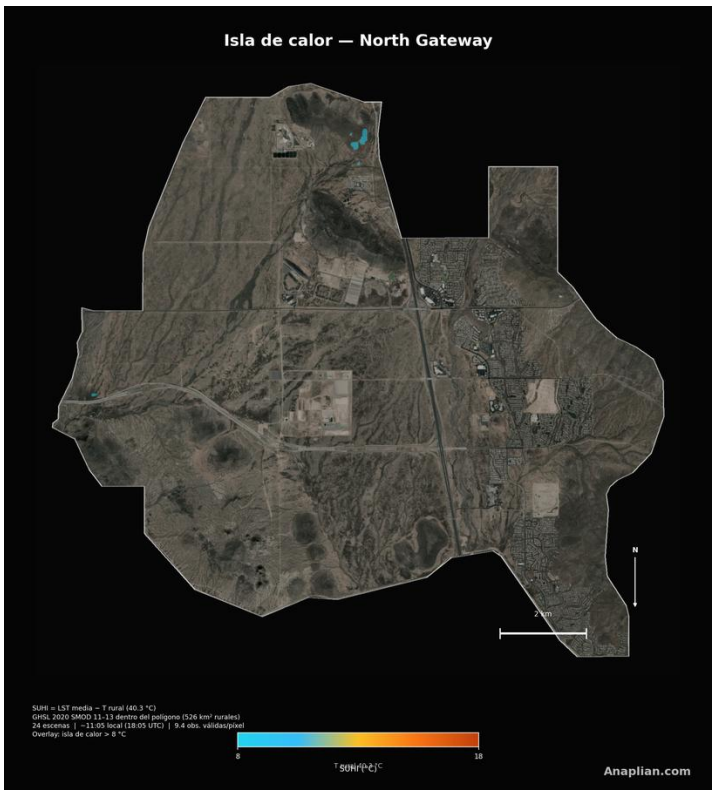
Peak LST > 43 C



NDVI



Mean LST > 35 C



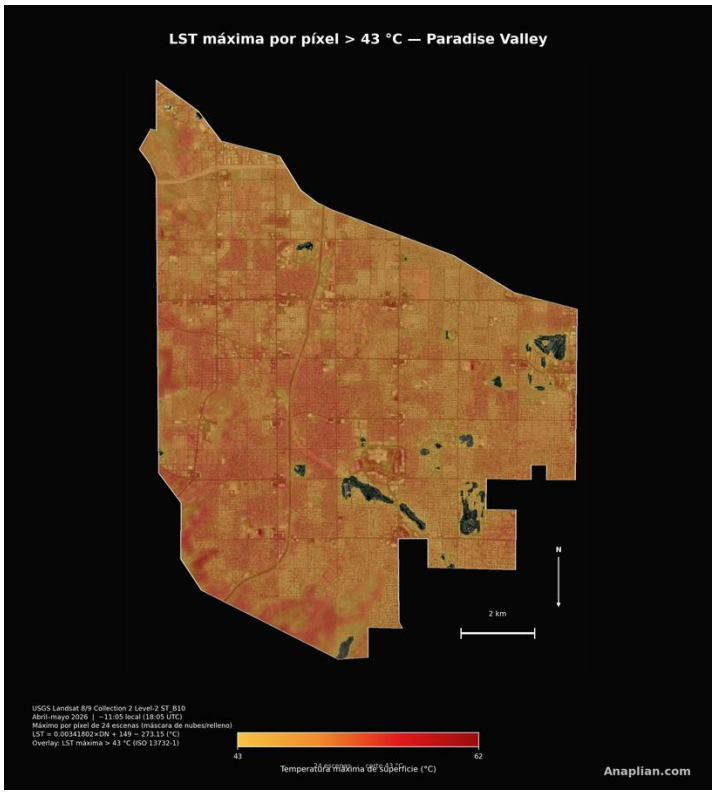
SUHI vs rural

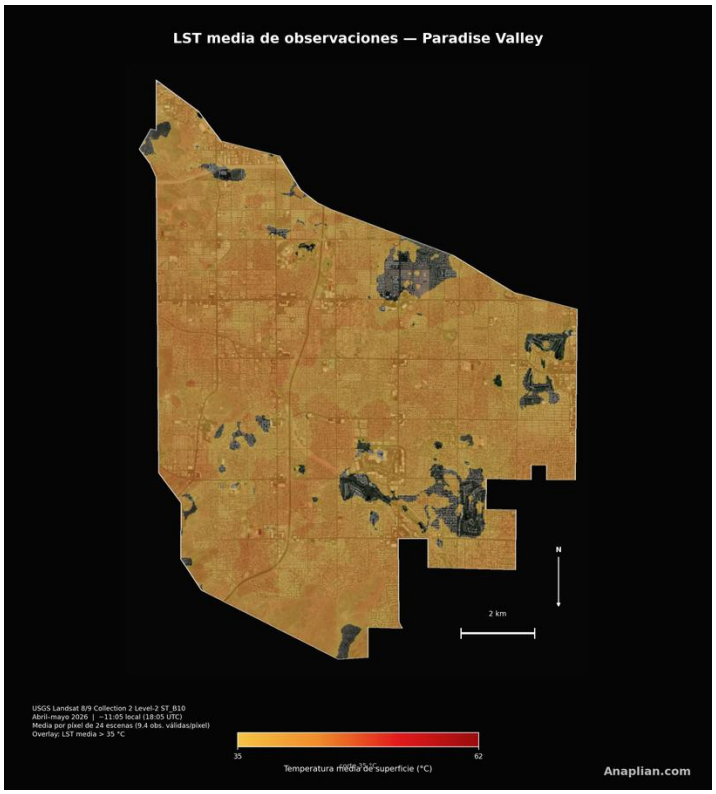
13. Paradise Valley · IVC 45.6

Larger lots (IVC 46). A little more canopy; 10 % of the mean > 43 °C. Pixel-max (98 %) shows tile roofs still crossing the threshold.

LST max > 43 C: 98% urban (49.2 C). LST mean > 43 C: 10% urban (39.5 C). SUHI -0.9 C vs T rural 40.3 C. NDVI 0.18; 105.0 km2; 8.0 obs./pixel.

On large lots: relatively more garden; the roof still crosses 43 °C. The mean (10 %) drops because there is less continuous asphalt.

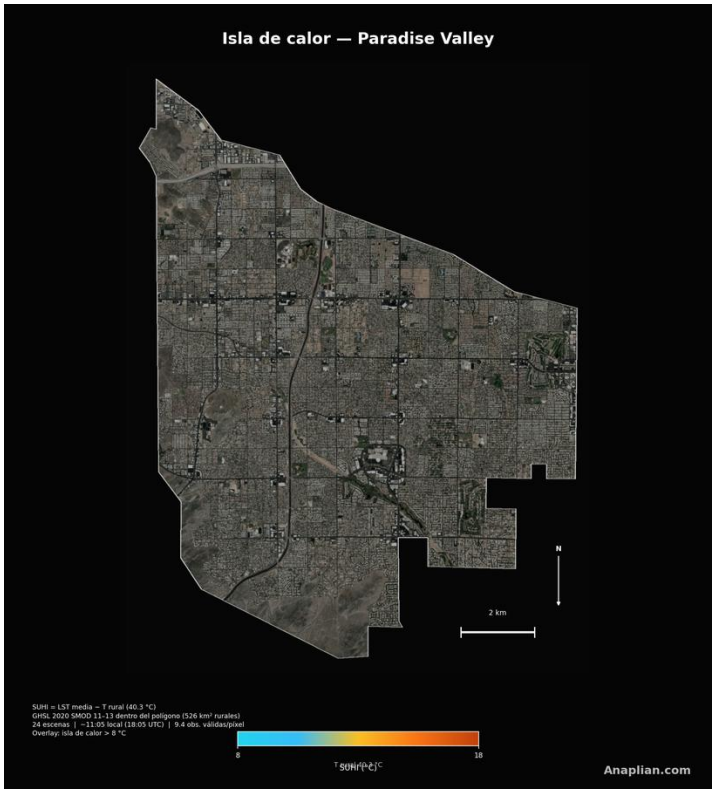




Peak LST > 43 C

Mean LST > 35 C





NDVI

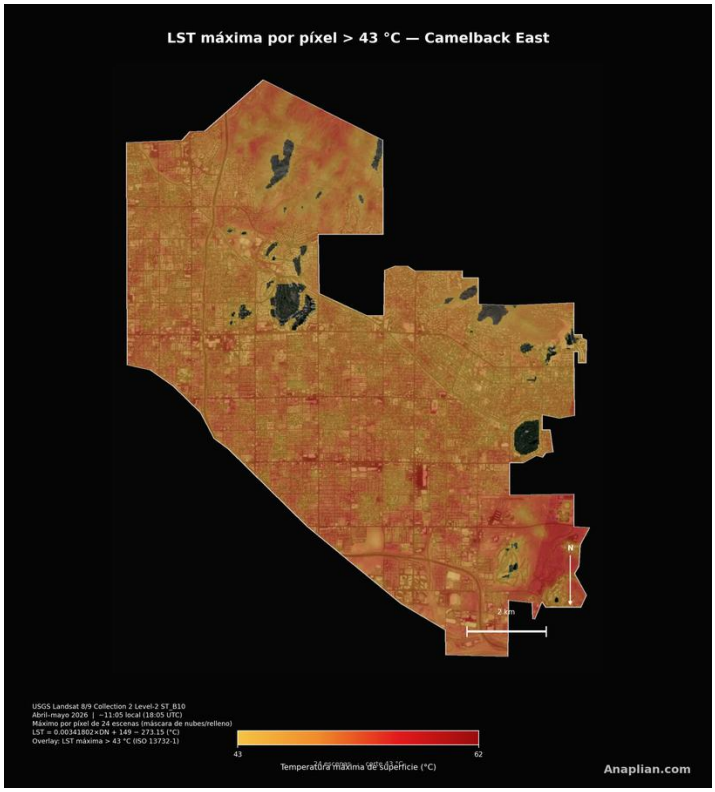
SUHI vs rural

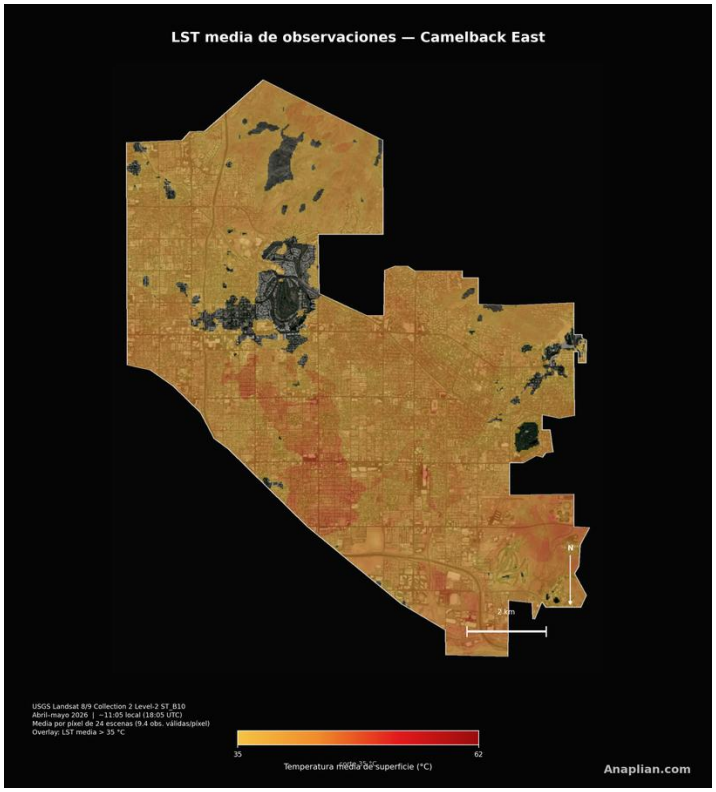
14. Camelback East · IVC 44.7

Camelback and avenues (IVC 45). NDVI 0.22, the highest in the eastern core; 16 % of the mean > 43 °C. Resorts and arterial: fragmented shade on asphalt.

LST max > 43 C: 97% urban (49.1 C). LST mean > 43 C: 16% urban (39.9 C). SUHI -0.5 C vs T rural 40.3 C. NDVI 0.22; 83.3 km2; 8.5 obs./pixel.

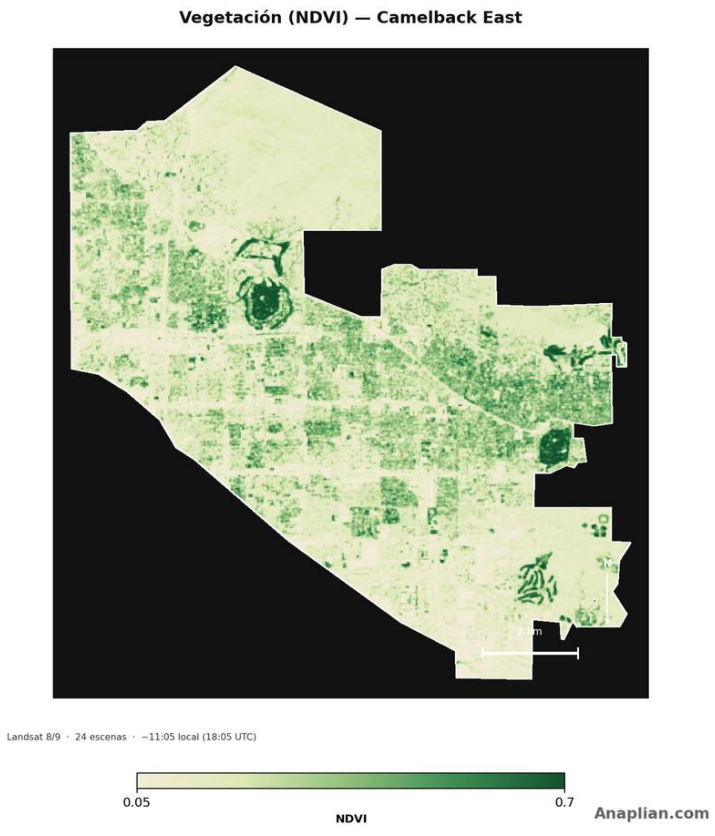
Under Camelback: resorts, avenue and uneven canopy. Greenness 0.22 is the eastern high; the corridor asphalt stays hot.

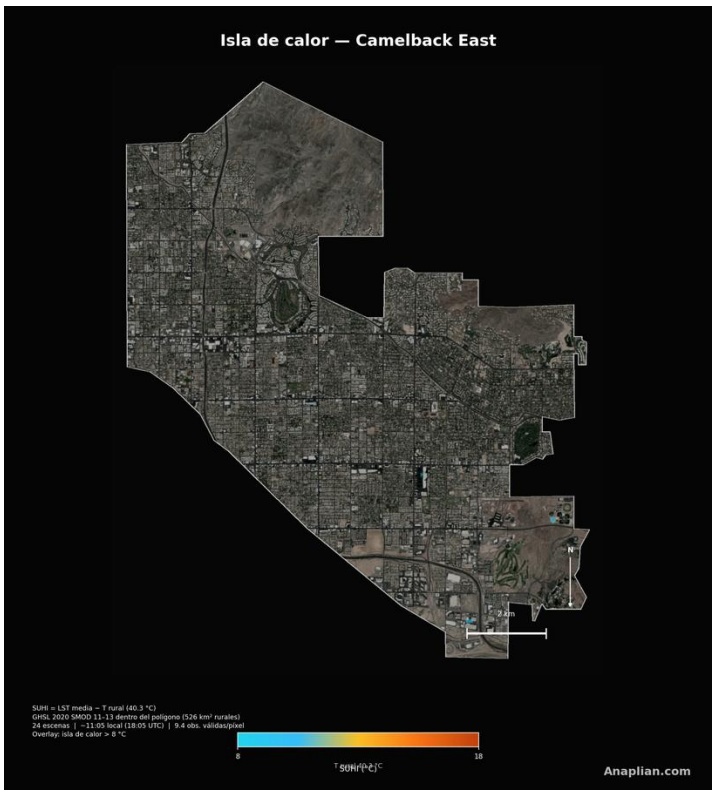




Peak LST > 43 C

Mean LST > 35 C





NDVI

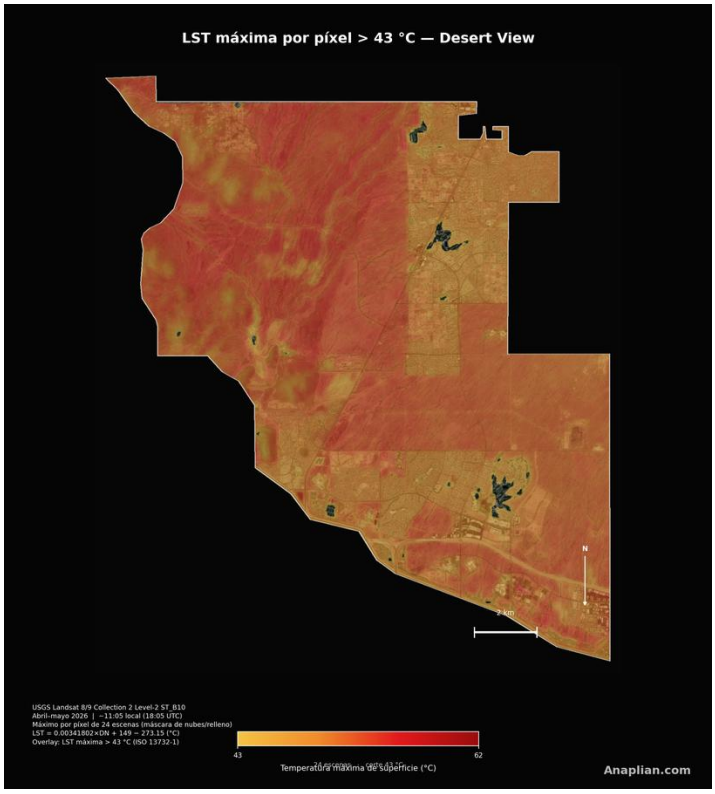
SUHI vs rural

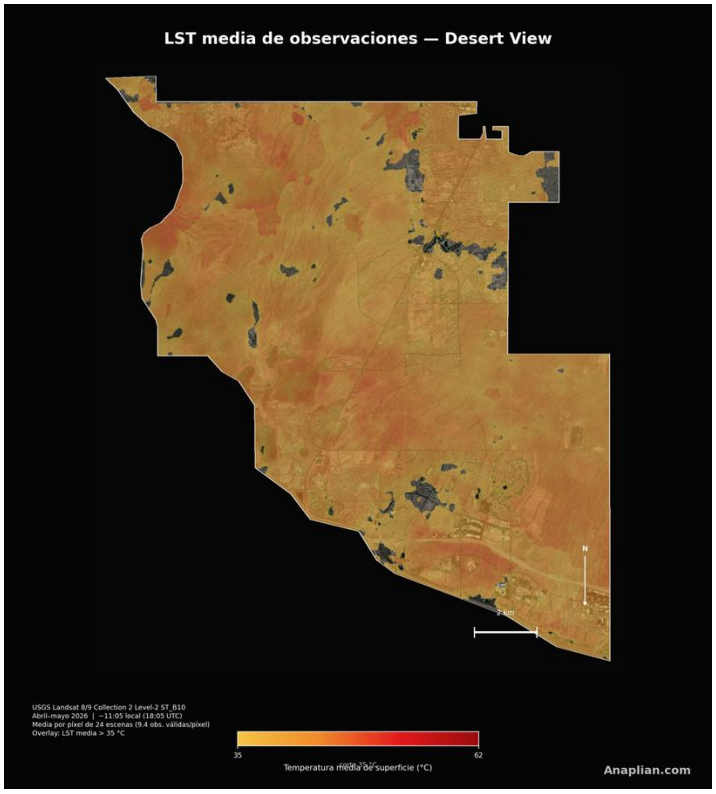
15. Desert View · IVC 43.6

The least vulnerable (IVC 44). Northern conservation land and low density; only 7 % of the urban mean > 43 °C. Pixel-max is still 98 %: desert does not forgive the roof, it does forgive the mean.

LST max > 43 C: 98% urban (48.8 C). LST mean > 43 C: 7% urban (39.1 C). SUHI -1.2 C vs T rural 40.3 C. NDVI 0.20; 51.7 km2; 13.4 obs./pixel.

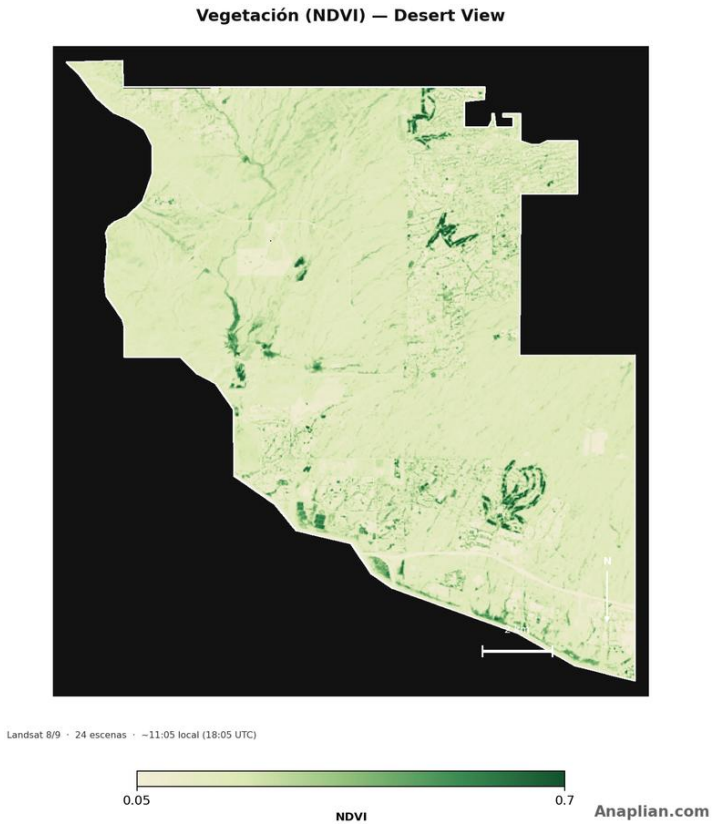
On northern conservation land: low density and 7 % of the mean > 43 °C. The April roof still runs red at pixel-max; the mean no longer does.





Peak LST > 43 C

Mean LST > 35 C





NDVI

SUHI vs rural

11. Why Phoenix is in Priority 2

Priority 2 follows the same cut — more than two million inhabitants in 18–42 °N — and the same hazard × exposure × vulnerability frame, one step below the six headline cities. It does not replace Delhi or Karachi: it complements them. Lahore, Dhaka and Ahmedabad concentrate the growth of extreme-heat person-days in South Asia (Tuholske 2021); Phoenix is the ECOSTRESS reference for pavement burns (49–60 °C) already used in the project's map style; Tehran adds a semi-arid megacity in a closed basin; Monterrey is Mexico City's dry-industrial counterpart.

The metric is the same ivc-l4-v1 and the window remains April–May 2026, so that an IVC of 60 in a Lahore town is comparable to Iztapalapa or to a Phoenix urban village. In Lahore, Ahmedabad and Dhaka that window is pre-monsoon or hot season. In Phoenix and Monterrey the annual peak is June–August; they enter in spring on purpose so as not to break the fixed scale. Ahmedabad also has a pioneering heat-action plan in India: the index measures surface exposure, not the plan's success.

Phoenix is the desert-pavement heat benchmark: urban LST of 49–60 °C in the core, with little canopy. The city's 15 urban villages show whether the core (Central City, Maryvale) runs hotter than the planned desert of the north (Desert View, North Gateway). April–May is not the July peak; it enters to anchor the scale.